A

Major Project Stage-I Report On

INTEGRATING DEEP LEARNING FOR EARLY DETECTION AND PREVENTION OF HEART DISEASE

(Submitted partial fulfillment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

By

M. Meghna (227R1A6230)

CH. Abhilash (227R1A6210)

P. Dinakar (227R1A6237)

Under the Guidance of

Mrs. E. Sushma

Associate Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CMR TECHNICAL CAMPUS

UGC AUTONOMOUS

(Accredited by NAAC, NBA, Permanently Affiliated to JNTUH, Approved by AICTE, New Delhi) Recognized Under Section 2(f) & 12(B) of the UGCAct.1956, Kandlakoya (V), Medchal Road, Hyderabad-501401.

November 2025.

DEPARTMENT OF COMUPTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CERTIFICATE

This is to certify that the Project entitled "**INTEGRATING DEEP LEARNING FOR EARLY DETECTION AND PREVENTION OF HEART DISEASE**" being submitted by **M. MEGHNA(227R1A6230), CH. ABHILASH (227R1A6210), P. DINAKAR (227R1A6237),** in partial fulfillment of the requirements for the award of the degree of B. Tech in COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY) to the Jawaharlal Nehru Technological University, Hyderabad is a record of bonafide work carried out by them under our guidance and supervision during Academic Year 2025 – 2026.

The result embodies in this thesis have not been submitted to any other University or Institute for the award of any other degree or diploma.

Mrs. E . SUSHMA	**Dr. K. MURALI**
Associate Professor	**Professor & HOD**
Internal Guide	

ACKNOWLEDGEMENT

We take this opportunity to express our gratitude to the people who have been instrumental in the successful completion of this project, we take this opportunity to express our profound gratitude and deep regard to our guide Dr. B. Mohan Babu, Designation for his exemplary guidance, monitoring and constant encouragement throughout the project work.

We also take this opportunity to express a deep sense of gratitude to the Project Review Committee (PRC) coordinators Mrs. V. Tejaswini, Mrs. K. Sudha Pavani for their cordial support, valuable information and guidance, which helped us in completing this task through various stages.

We are also thankful to Dr. K. Murali, Head, Department of Computer Science and Engineering (Data Science) for providing encouragement and support for completing this project successfully.

We are deeply grateful to Dr. A. Raji Reddy, Director, for his cooperation throughout the course of this project. Additionally, we extend our profound gratitude to Sri. Ch. Gopal Reddy, Chairman, Smt. C. Vasantha Latha, Secretary and Sri. C. Abhinav Reddy, Vice Chairman, for fostering an excellent infrastructure and a conducive learning environment that greatly contributed to our progress.

The guidance and support received from all the members of CMR Technical Campus who contributed to the completion of the project. We are grateful for their constant support and help.

Finally, we would like to take this opportunity to thank our family for their constant encouragement, without which this assignment would not be completed. We sincerely acknowledge and thank all those who gave support directly and indirectly in the completion of this project.

M. MEGHNA (227R1A6230)

CH. ABHILASH (227R1A6210)

P. DINAKR (227R1A6237)

ABSTRACT

Diabetes is the ‘‘mother of all diseases’’ as it affects multiple organs of body of an individual in some way. Its timely detection and management are critically important. Otherwise, the long run, it can cause several complications in a diabetic. Heart disease is one of the major complications of diabetes. This work proposed an Optimal Scrutiny Boosted Graph Convolutional LSTM (O-SBGC-LSTM), SBGC-LSTM enhanced by Eurygaster Optimization Algorithm (EOA) to tune hyper parameters for early prevention and detection of diabetes disease. This work proposed an Optimal Scrutiny Boosted Graph Convolutional LSTM (O-SBGC-LSTM), SBGC-LSTM enhanced by Eurygaster Optimization Algorithm (EOA) to tune hyper parameters for early prevention and detection of diabetes disease. This method not only captures discriminative features in spatial configuration and temporal dynamics but also explore the co-occurrence relationship between spatial and temporal domains. The top SBGC-LSTM layer's temporal receptive fields are expanded using this method's temporal hierarchical architecture, which also greatly lowers computation costs and improves the capacity to learn high-level semantic representation. Overal LSTM's performance was deemed satisfactory, with the majority of studies reporting accuracy. In practically every study that reported such comparison results, the suggested hybrid DL outperformed traditional machine learning techniques. Moreover, prevention is preferable to treatment. Utilizing suggestion tables, fuzzy-based inference techniques were also used to improve the prevention process.

TABLE OF CONTENTS

Certificate ii

Abstract iv

Table of Contents v

1. **INTRODUCTION** 1
    - 1.1.Scope Of the Project 2
    - 1.2.Objective 3
    - 1.3.Problem Statement 3
    - 1.4.Existing System 4
2. **LITERATURE REVIEW** 5
    - 2.1.Literature Review 5
    - 2.2.System Requirements 7
3. **PROJECT DESCRIPTION** 9
    - 3.1.General 9
    - 3.2.Methodologies 9
4. **SYSTEM DESIGN** 17
    - 4.1.Use Case Diagram 17
    - 4.2.Class Diagram 19
    - 4.3.Object Diagram 20
    - 4.4.State Chart Diagram 21
    - 4.5.Sequenced Diagram 22
    - 4.6.Collaboration Diagram 23
    - 4.7.Activity Diagram 24
    - 4.8.Component Diagram 25
    - 4.9.E-R Diagram 26
    - 4.10.Data Flow Diagram 28
    - 4.11.Deployment Diagram 29
    - 4.12.System Architecture 30
5. **IMPLEMENTATION** 31

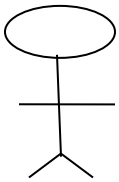
5.1.Administration Authentication 31
5.2. Doctor Registration Functionality 33
5.3. Doctor Login Authentication 36
6. **TESTING AND DEBUGGING** 38
6.1.Developing Methodologies 38
6.2. Unit testing 38
6.3.Functional test 39
6.4.System Test 39
6.5.Performance Test 39
7. **RESULT AND DISCUSSION** 40
7.1.Various Snapshots 41
8. **CONCLUSION** 46
9. **BIBLIOGRAPHY** 47

LIST OF FIGURES

FIGURE NO	NAME OF THE FIGURE	PAGE.NO
1.	SMART ART OF THE PROPOSED METHOLOGY	2
2.	USER INTERFACE	11
3.	ADMIN LOGIN FLOWCHART	12
4.	USER REGISTRATION FLOWCHART	13
5.	HOSPITAL LOGIN FLOWCHART	14
6.	DOCTOR LOGIN FLOWCHART	15
7.	LAB TECHNICIAN LOGIN FLOWCHART	16
8.	HOME PAGE	40
9.	ADMIN LOGIN	40
10.	ADD DOCTOR	41
11.	DOCTOR LIST	41
12.	USER DETAILS	42
13.	PREDICTION OF DISEASE	42

14.	HOSPITAL LOGIN	43
15.	PATIENT ADMISSION	43
16.	DOCTORS LOGIN	44
17.	USER REGISTRATION	44
18.	USER LOGIN	45
19.	LAB PERSON LOGIN FOR PREDICTION DETAILS	45

LIST OF NOTATIONS

S.NO	NAME	NOTATION	DESCRIPTION
1.	Class	*+ public* *-private* *Class Name* *-Attribute-attribute* *+operation* *+operation*	Represents a collection of similar entities grouped together.
2.	Association	NAM Class A — Class B Class A — Class B	Associations represents static relationships between classes. Roles represents the way the two classes see each other.
3.	Actor		It aggregates several classes into a single classis

4.	Aggregation	Class A ↑ Class B Class A ↑ Class B	Interaction between the system and external environment
5.	Relation (uses)	Uses	Used for additional process communication.
6.	Relation (extends)	extends →	Extends relationship is used when one use case is similar to another use case but does a bit more.
7.	Communication	———	Communication between various use cases.
8.	State	State	State of the process.
9.	Initial State		Initial state of the object
10.	Final state		Final state of the object

11.	Control flow		Represents various control flow between the states.
12.	Decision box		Represents decision making process from a constraint
13.	Use case	Uses case	Interact ion between the system and external environment.
14.	Component		Represents physical modules which is a collection of components.
15.	Node		Represents physical modules which are a collection of components.
16.	Data Process / State		A circle in DFD represents a state or process which has been triggered due to some event or action.
17.	External entity		Represents external entities such as keyboard, sensors, etc.

18.	Transition		Represents communication that occurs between processes.
19.	Object Lifeline		Represents the vertical dimensions that the object communications.
20.	Message	Message	Represents the message exchanged.

LIST OF ABBREVATION

S.NO	ABBREVATION	EXPANSION
1.	DB	Data Base
2.	JVM	Java Virtual Machine
3.	JSP	Java Server Page
4.	CB	Collective Behaviour
5.	RSSS	Ramp secret sharing scheme
6.	JRE	Java Runtime Environment

1. INTRODUCTION

1 INTRODUCTION

The incidence of diabetes has increased over the last decade, posing a new challenge to India's health policymakers. According to the International Diabetes Federation (IDF), approximately 537 million people across the globe were living with diabetes in 2021. The IDF has also projected that this number will increase to 643 million by 2030 [1]. By 2035, Indian diabetic counts are expected to exceed 80 million, making it "Diabetes Capital" of the world according to the Diabetes Foundation of India [2]. Diabetes prevalence increased in both rural and urban regions of India between 2005 and 2015 according to an analysis of (National Family Health Survey data) [3].

Over nutrition has been found to be a major risk factor for a number of diseases such as diabetes, hypertension, heart diseases, certain type of cancers, etc. [4], [5]. Overnutrition is one of the potential factors that may generate insulin resistance, which in turn may increase the sugar or glucose content in the blood leading to diabetes [6]. The study [7], finds similar results among married women in Delhi, India. Other factors that may lead to diabetes include smoking, alcohol consumption, high sugar intake, genetic predisposition, etc [8]. Type 2 diabetes mellitus (T2DM) can result in a number of complications, such as macrovascular diseases, for example, cardiovascular disease (CHD), and microvascular diseases, for example, kidney, the retina failure and the nervous system diseases [9]. Even worse, T2DM may cause dementia and cognitive impairment, thereby reducing sensitivity to diabetes complications for T2DM patients. It is known that the incidence of heart disease such as Heart Failure (HF), cardiac dysfunction in individuals with T2DM is much higher than those without T2DM. Specifically, coronary heart disease (CHD) represents one of the most common and severe diabetes complications. Since CHDs have high incidence and fatality rates, it is critical for people to be aware of CHDs. Diabetics have a two to four times higher risk of developing CHDs, and 68% of diabetics older than 65 years with CHDs have died. Numerous approaches, including mathematical models such as Cox regressions and Machine Learning Techniques (MLTs) such as neural networks, have been used to predict CHDs [10], [11].

As shown in Fig 1, this work presented an artificial intelligence-based approach to predict the probability of developing CHDs in persons with T2DM.

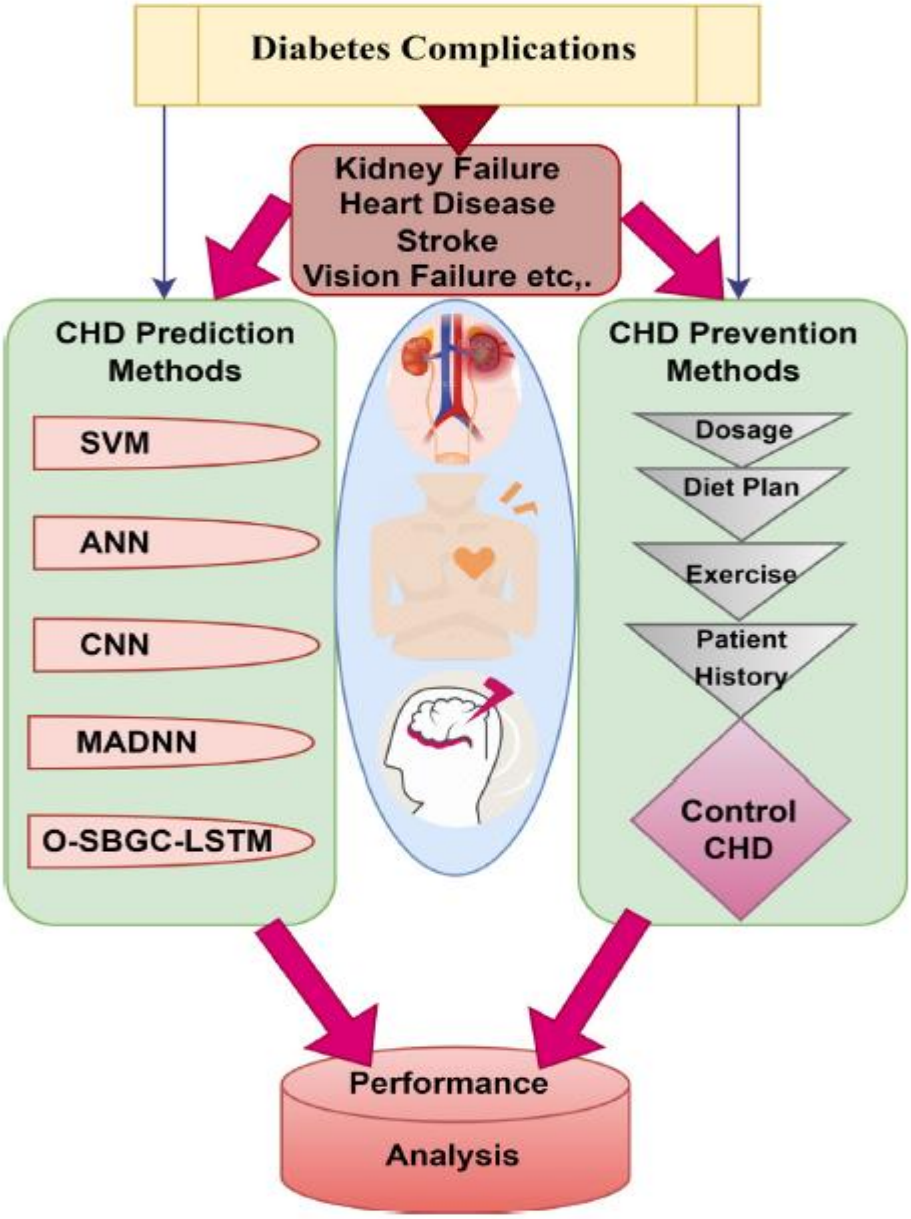


FIGURE 1. Smart Art of the proposed methodology.

1.1Scope of the Project

The project focuses on developing an intelligent deep-learning framework for the early detection and prevention of Coronary Heart Disease (CHD) among diabetic patients. The system aims to assist doctors, hospitals, and patients by predicting CHD risk levels based on

patient history, test results, and lifestyle factors. It also provides preventive recommendations such as dietary adjustments, exercise guidance, and medication management.. This project contributes to the healthcare sector by combining AI-based predictive analytics and fuzzy-logic-driven preventive guidance for effective monitoring and management of diabetic heart-disease risk.

1.2 Objective

The main objectives of the project are as follows:

1. To design and implement a hybrid deep-learning model (O-SBGC-LSTM) for accurate prediction of CHD in diabetic patients.

2. To optimize model performance using the Eurygaster Optimization Algorithm (EOA) for fine-tuning hyperparameters.

3. To build a web-enabled platform that allows patients, doctors, and hospitals to share and analyze clinical data seamlessly.

4. To employ fuzzy-inference-based preventive mechanisms that recommend lifestyle and medical interventions to reduce CHD risk.

5. To enhance early diagnosis accuracy, minimize manual errors, and improve decision-making efficiency in real-time clinical environments.

1.3 PROBLEM STATEMENT

- Deep learning-based model to predict the risk of developing CHD for individuals with T2DM. The data is pre-processed, and then feature extraction and classification are achieved O-SBGC-LSTM classifier to diagnose disease. The EOA is combined with SBGC-LSTM to create the O-SBGC-LSTM.

- After detecting CHD in diabetic patient with the help of O-SBGC-LSTM classifier, to control the disease by applying prevention methods like diet control, fitness of the body, dosage of medicine, and patient medical history.

1.4 EXISTING SYSTEM

- Diabetes is the "mother of all diseases" as it affects multiple organs of body of an individual in some way. Its timely detection and management are critically important. Otherwise, the long run, it can cause several complications in a diabetic.

- Heart disease is one of the major complications of diabetes. This work proposed an Optimal Scrutiny Boosted Graph Convolutional LSTM (O-SBGC-LSTM), SBGC-LSTM enhanced by Eurygaster Optimization Algorithm (EOA) to tune hyper parameters for early prevention and detection of diabetes disease.

- This work proposed an Optimal Scrutiny Boosted Graph Convolutional LSTM (O-SBGC-LSTM), SBGC-LSTM enhanced by Eurygaster Optimization Algorithm (EOA) to tune hyper parameters for early prevention and detection of diabetes disease.

- This method not only captures discriminative features in spatial configuration and temporal dynamics but also explore the co-occurrence relationship between spatial and temporal domains.

1.4.1EXISTING SYSTEM DISADVANTAGES

- Computing makes the environment more complex.
- The current computing architecture has the problems.
- The edge node carries too many task requests.

2. LITERATURE REVIEW

LITERATURE REVIEW

2.1 LITERATURE REVIEW

Title: Heterogeneity of dietary practices in India: Current status and implications for the prevention and control of type 2 diabetes

Year: 2023

Author: M. Sachdev and A. Misra.

Description: Diet is one of the important modifiable factors in prevention of type 2 diabetes (T2D), making it important to understand geographical variations of food consumption pattern, their similarities and differences across various regions. Asian Indian diet patterns are mostly carbohydrate-based and with vast regional diversity. Staple food, food groups consumed, meat consumption pattern, type of fermented foods, food preparations, type of cooking oil used and food ingredients vary with different regions of India. There has been a slow transition from consumption of coarse grains to refined grains owing to socio-economic, cultural and other factors. Consumption of fruits and vegetables, fats, ready-to-eat foods and sugar is higher in urban population whereas consumption of carbohydrates in the form of cereals and millets is higher among the rural population. Cereal grains followed by pulses and legumes are main sources of protein, given that the frequency of meat consumption is low even among non-vegetarians in India compared to other countries. Overall, there is a tendency towards consumption of calorie-dense foods at the cost of food diversity, which may also result in micronutrient deficiencies as well as development of T2D and related metabolic diseases. Public health strategies and policy level decisions involving stakeholders with diet and lifestyle modification as focal points are absolute priorities to prevent and manage the burden of obesity and T2D in India.

Title: A deep learning approach to predict diabetes' cardiovascular complications from administrative claims

Year: 2021

Author: E. Longato, G. P. Fadini, G. Sparacino, A. Avogaro, L. Tramontan and B. Di Camillo

Description: People with diabetes require lifelong access to healthcare services to delay the onset of complications. Their disease management processes generate great volumes of data across several domains, from clinical to administrative. Difficulties in accessing and processing these data hinder their secondary use in an institutional setting, even for highly desirable applications, such as the prediction of cardiovascular disease, the main driver of excess mortality in diabetes. Hence, in the present work, we propose a deep learning model for the prediction of major adverse cardiovascular events (MACE), developed and validated using the administrative claims of 214,676 diabetic patients of the Veneto region, in North East Italy. Specifically, we use a year of pharmacy and hospitalisation claims, together with basic patient's information, to predict the 4P-MACE composite endpoint, i.e., the first occurrence of death, heart failure, myocardial infarction, or stroke, with a variable prediction horizon of 1 to 5 years. Adapting to the time-to-event nature of this task, we cast our problem as a multi-outcome (4P-MACE and components), multi-label (1 to 5 years) classification task with a custom loss to account for the effect of censoring. Our model, purposefully specified to minimise data preparation costs, exhibits satisfactory performance in predicting 4P-MACE at all prediction horizons: AUROC from 0.812 (C.I.: 0.797 - 0.827) to 0.792 (C.I.: 0.781 - 0.802); C-index from 0.802 (C.I.: 0.788 - 0.816) to 0.770 (C.I.: 0.761 - 0.779). Components' prediction performance is also adequate, ranging from death's 0.877 1-year AUROC to stroke's 0.689 5-year AUROC.

Title: An intelligent fuzzy inference rule-based expert recommendation system for predictive diabetes diagnosis

Year: 2022

Author: P. Nagaraj and P. Deepalakshmi.

Description: Diabetes is one of the most common and hazardous diseases, which can affect almost every organ in the body. Diagnosis of diabetes requires determining all vital parameters related to the disease. However, the nature of the data from those parameters is very uncertain, affecting the process of disease diagnosis. This article proposes an intelligent fuzzy inference rule-based predictive diabetes diagnosis model (IFIR_PDDM), providing content recommendations to patients with diabetes. The suggested model employs an inference technique that medical specialists have validated for recommendations. IFIR_PDDM comprises three elements used to forecast the risk of diabetes disease. Initially, a fuzzy membership function utilizes medical recommendations and statistical methodologies. Medical specialists then validate the mining-based rules using a decision tree rule induction technique. The proposed model predicts the risk of diabetes disease using fuzzy inference based on Mamdani's technique. Based on this information, the recommendations for a normal life, nutrition, exercise, and medications are given to patients. We used an electronic health record (EHR) medical and clinical dataset from the PIMA Indian Diabetes dataset to develop our proposed model and assess its performance. The proposed model takes less time for diabetes diagnosis, and the expert recommendation system uses the fuzzy inference method.

2.2. SYSTEM REQUIREMENTS

2.2.1 HARDWARE REQUIREMENTS

The hardware requirements may serve as the basis for a contract for the implementation of the system and should therefore be a complete and consistent specification of the whole system. Software engineers use them as the starting point for the system design. It shoulds what the system and not how it should be implemented.

- PROCESSOR : PENTIUM IV 2.6 GHz, Intel Core 2 Duo.
- RAM : 512 MB DD RAM
- MONITOR : 15” COLOR
- HARD DISK : 40 GB

2.2.2 SOFTWARE REQUIREMENTS

The software requirements document is the specification of the system. It should include both a definition and a specification of requirements. It is a set of what the system should do rather than how it should do it. The software requirements provide a basis for creating the software requirements specification. It is useful in estimating cost, planning team activities, performing tasks, tracking the teams, and tracking the team’s progress throughout the development activity.

- Front End : J2EE (JSP, SERVLET)
- Back End : MY SQL 5.5
- Operating System : Windows 7
- IDE : Eclipse

3. PROJECT DESCRIPTION

3.1 GENERAL

The data were available from different databases and were not similar, but fuzzy logic handled the data. Fuzzy logic introduces the term linguistic for making decisions for these records. Linguistic follows between 0 and 1 values. This work uses trapezoidal fuzzy membership, and a certain degree is considered for all the fuzzy rules that are accessible. Here, corresponding vague rules for disease control were generated.

3.2 METHODOLOGIES

3.2.1 MODULES NAME:

This project having the following 5 modules:

- **User Interface Design**
- **Admin**
- **User**
- **Hospital**
- **Doctor**
- **Lab Technician**

3.2.2 MODULES EXPLANATION AND DIAGRAM

- **User Interface Design**

To connect with server user must give their username and password then only they can able to connect the server. If the user already exits directly can login into the server else, user must register their details such as username, password, Email id, City and Country into the server. Database will create the account for the entire user to maintain upload and download rate. Name will be set as user id. Logging in is usually used to enter a specific page. It will search the query and display the query.

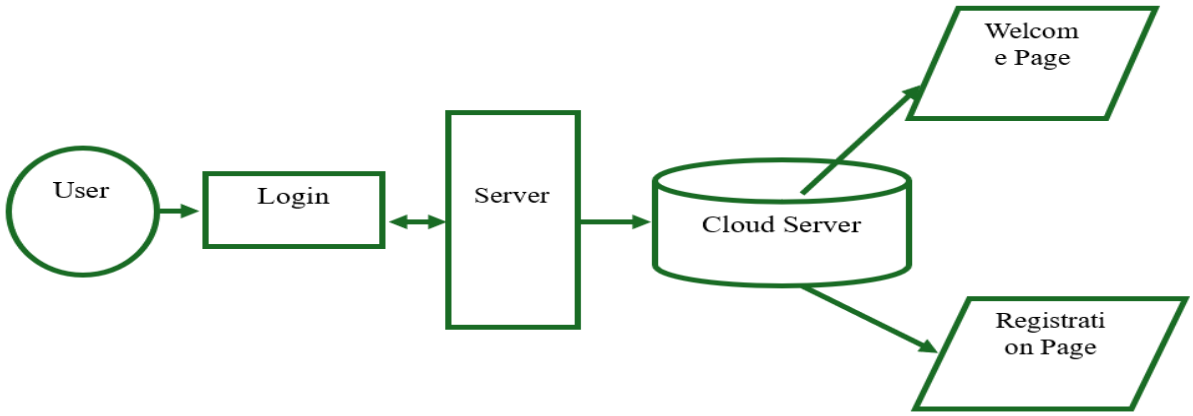


FIGURE 2:USER INTERFACE FLOWCHART

- **Admin**

This is the second module of our project no registration for admin only for login with help of admin name & password. Dataset is a collection of related sets of information that is composed of separate elements. Admin having some options like Add Doctor Details, View User Details, View Feedback, View Doc Details & View Test reports also.

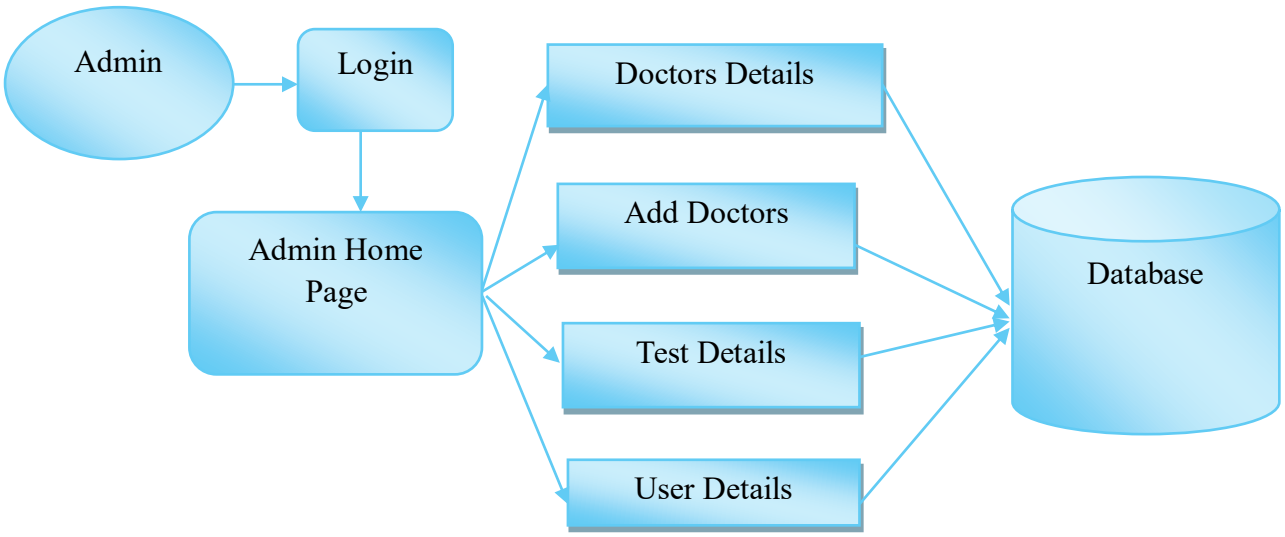


FIGURE 3: ADMIN LOGIN FLOWCHART

- **User**
  - Register (With Details like Age, Sex, etc.)
  - Login user then send to symptoms to hospital then get appointment to doctor.
  - Take the required tests provided by the doctor.
  - Based on the reports know the risk level then if risk is high then admit in hospital or if risk is low take the precautions and medicine provided by doctor.
  - System will accordingly view Doctor to consult.
  - Give Feedback & View Doctors

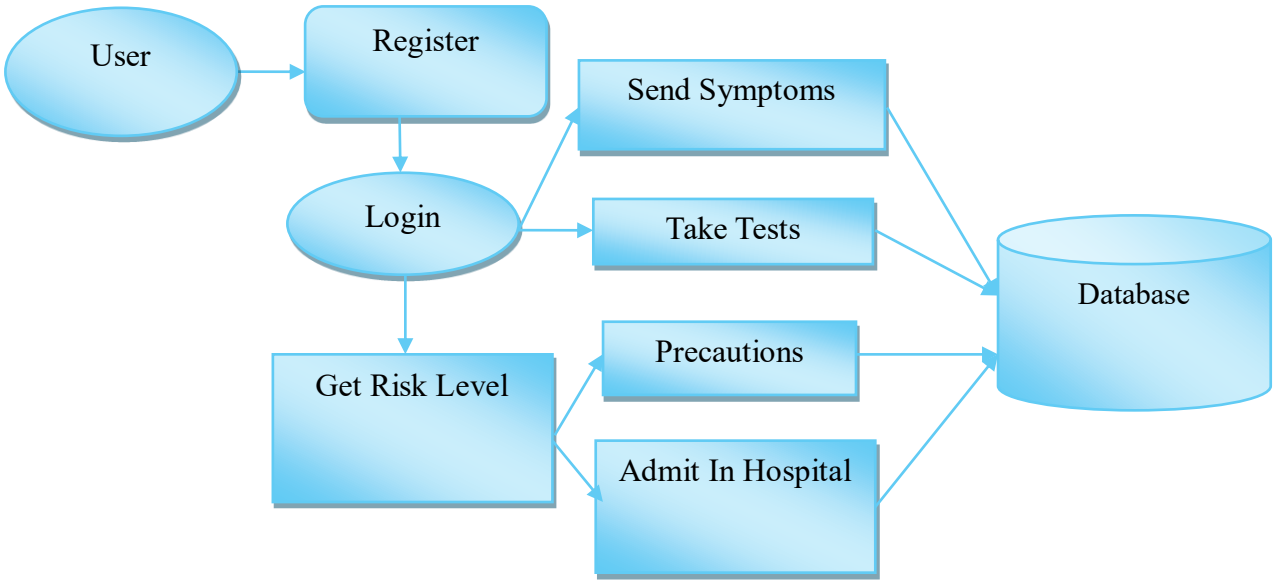
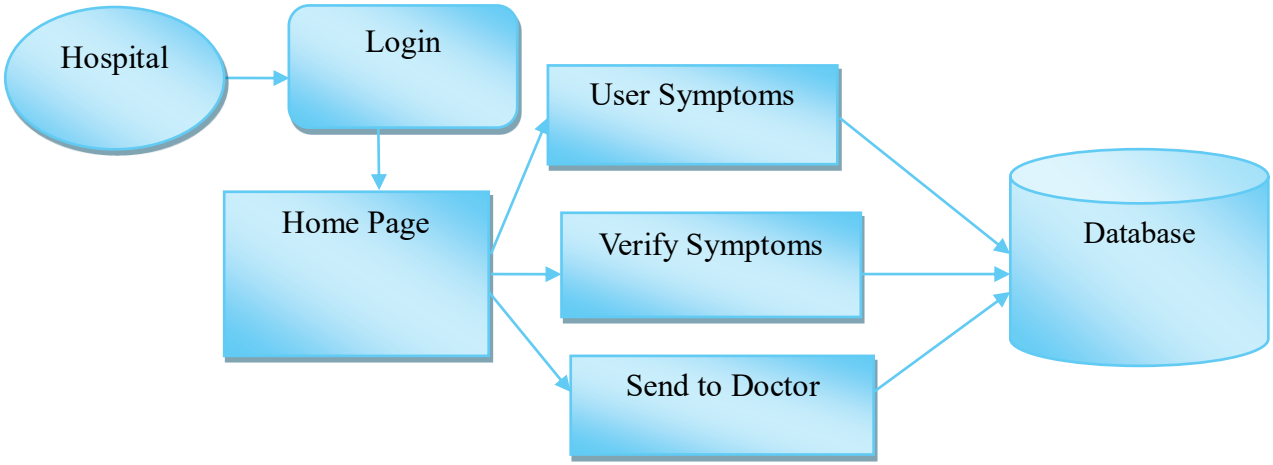


FIGURE 4: USER REGISTRATION FLOWCHART

➢ Hospital

This is the fourth module of our project no registration for hospital only for login with help of hospital name & password. After login verify user's symptoms then provide appointment with doctor. Then according to doctors reports if user wants to admit then provide the admission in the hospital otherwise provide medicine details. And take care of details, reports, and admission details of the user.



FIGIURE 5: HOSPITAL LOGIN FLOWCHART

➢ Doctor

This is fifth module in our project in this module based on doctor information it works. Admin already adding some doctors based on name & password login in to application. Then First doctor register in administration staff then the particular doctor authorized to this organization. After that every time login in to hospital then checks the patient's condition and doctor checks patient medical reports. Then check the risk level and then based on the risk level doctor will provide some suggestions to hospital staff (either admission or medication details).

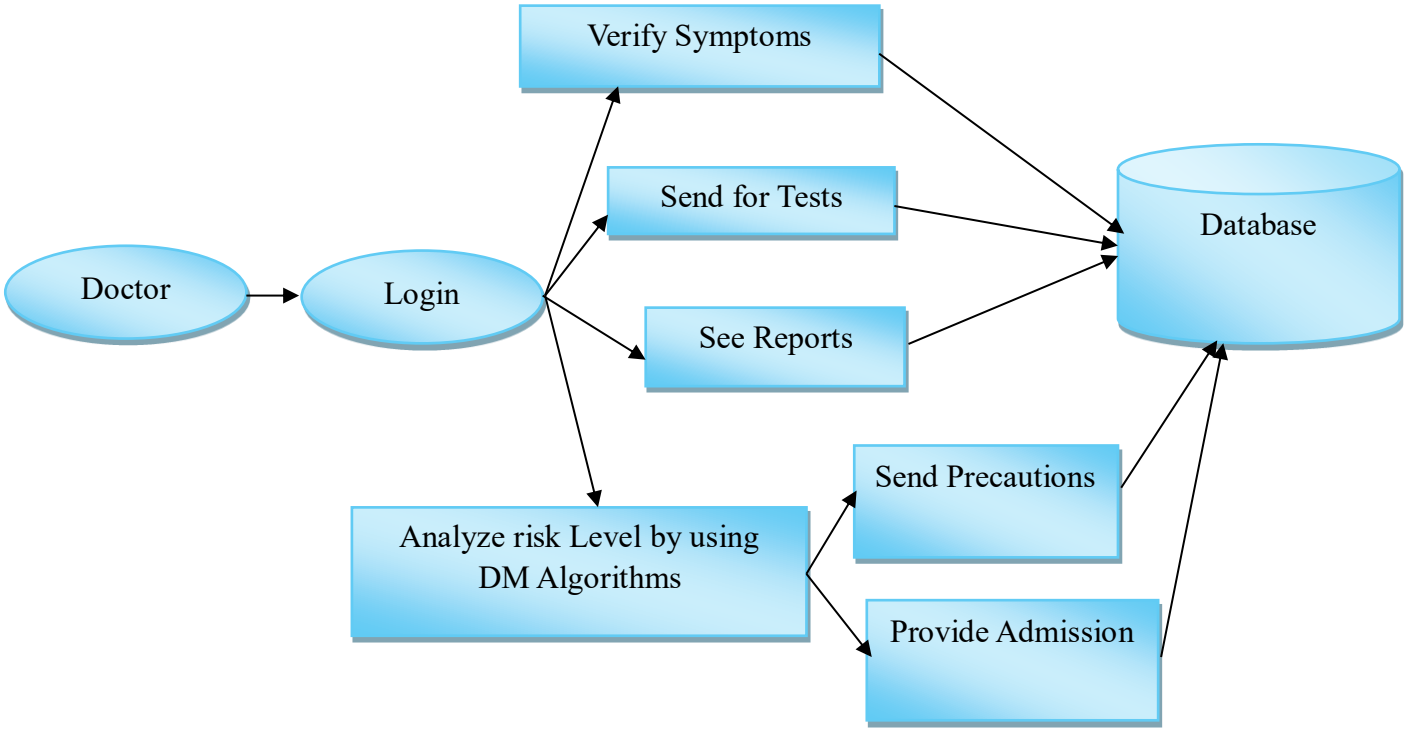


FIGURE 6: DOCTOR LOGIN FLOWCHART

➢ Lab Technician

This is the final module of our project after login lab technicians might collect samples, study and perform tests on body fluids, teeth, chemical compounds, biological specimens, or other fields of science. Lab techs use various types of machinery, lab equipment and complex computer programs to perform their tests. Lab techs also record their data or findings for study and scrutiny by doctors and physicians. Sending the attribute values & view the tests to conduct user.

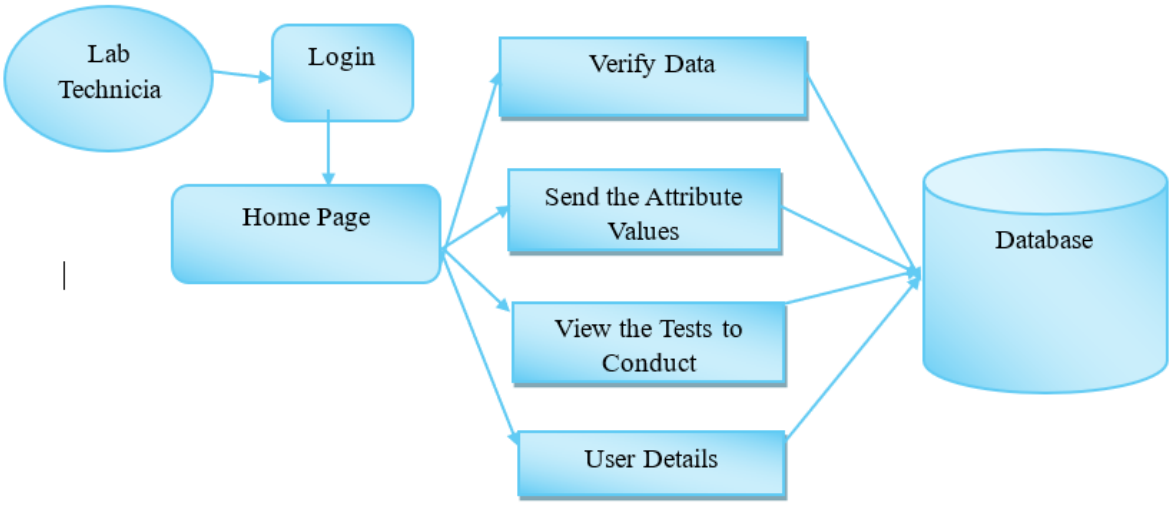


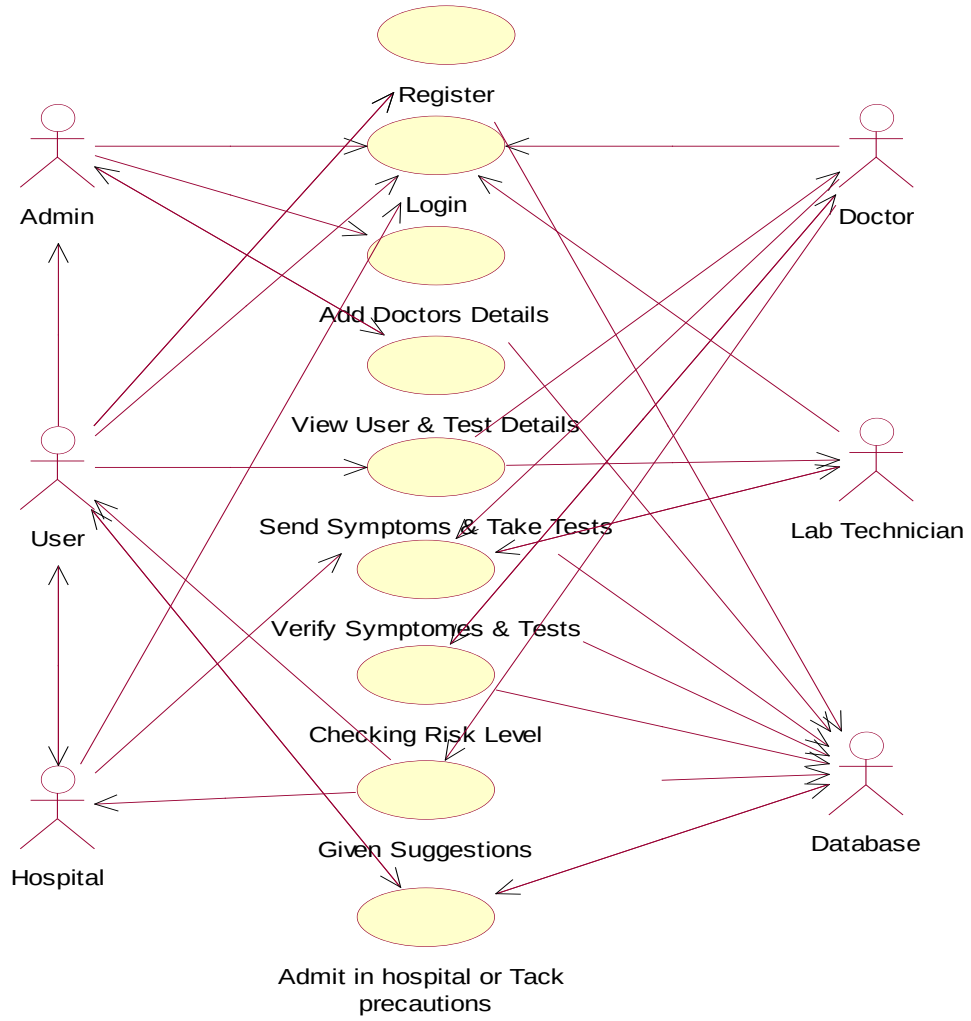
FIGURE 7: LAB TECHNICIAN LOGIN FLOWCHART

4. SYSTEM ARCHITECTURE & DESIGN

4.SYSTEM ARCHITECTURE & DESIGN

Design Engineering deals with the various UML [Unified Modelling language] diagrams for the implementation of project. Design is a meaningful engineering representation of a thing that is to be built. Software design is a process through which the requirements are translated into representation of the software. Design is the place where quality is rendered in software engineering. Design is the means to accurately translate customer requirements into finished product.

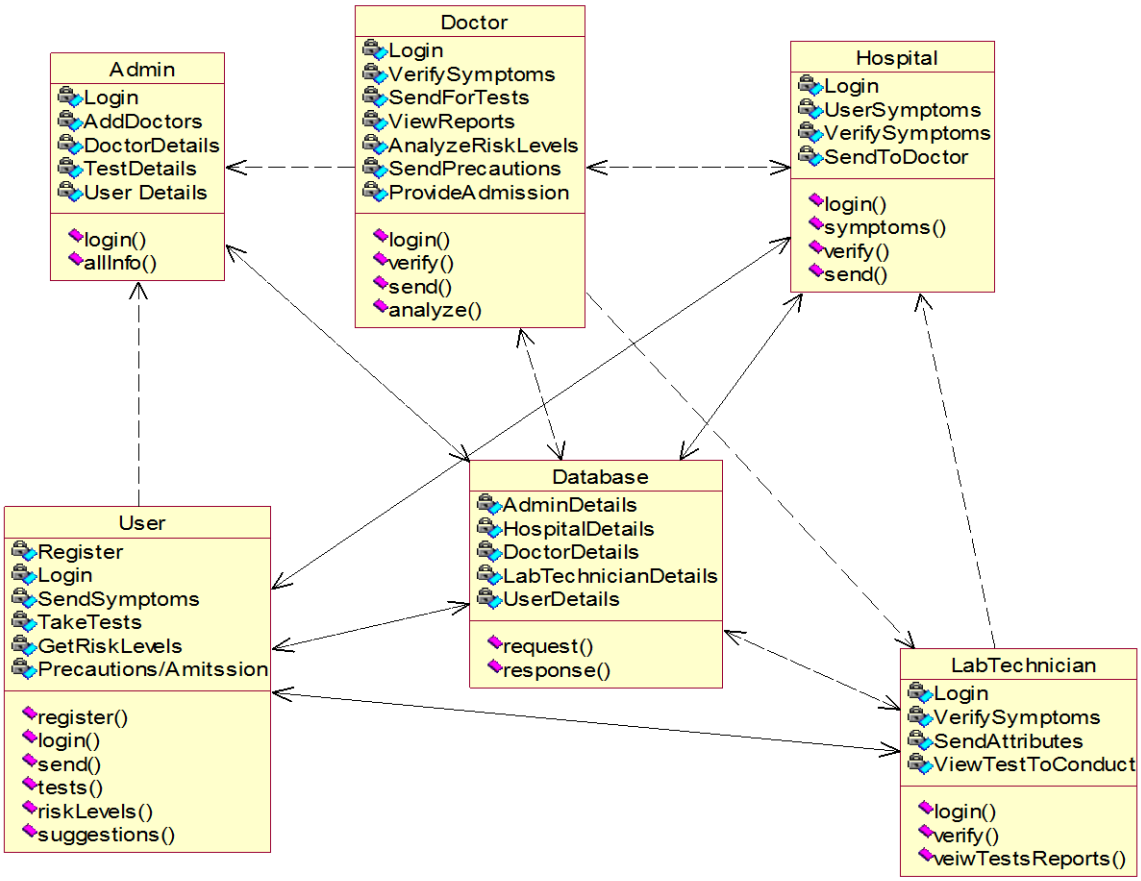
4.1 Use Case Diagram



EXPLANATION:

The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted. The above diagram consists of user as actor. Each will play a certain role to achieve the concept.

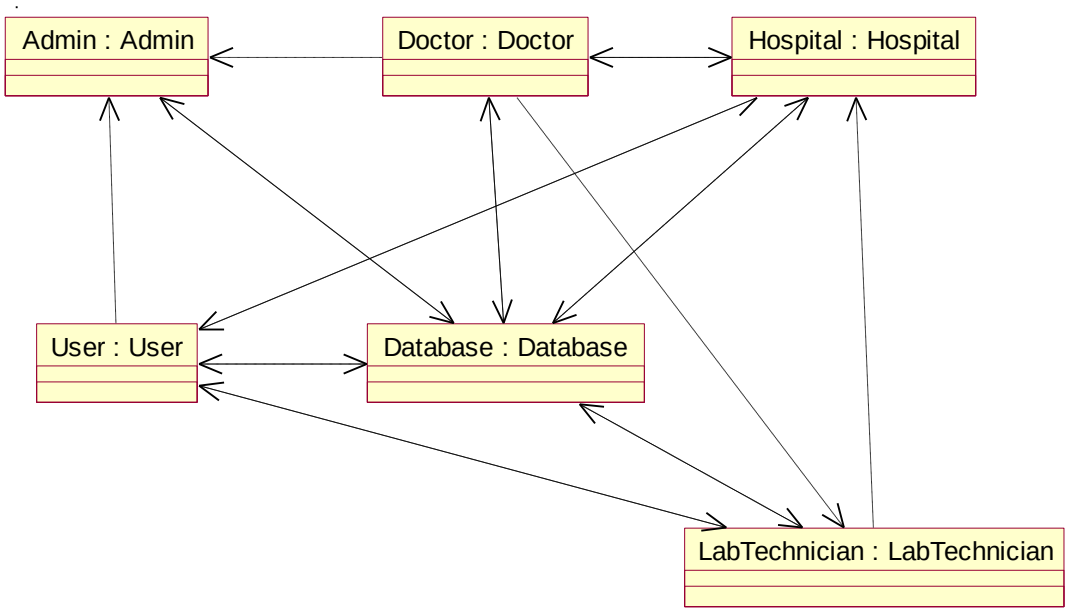
4.2 Class Diagram



EXPLANATION

In this class diagram represents how the classes with attributes and methods are linked together to perform the verification with security. From the above diagram shown the various classes involved in our project

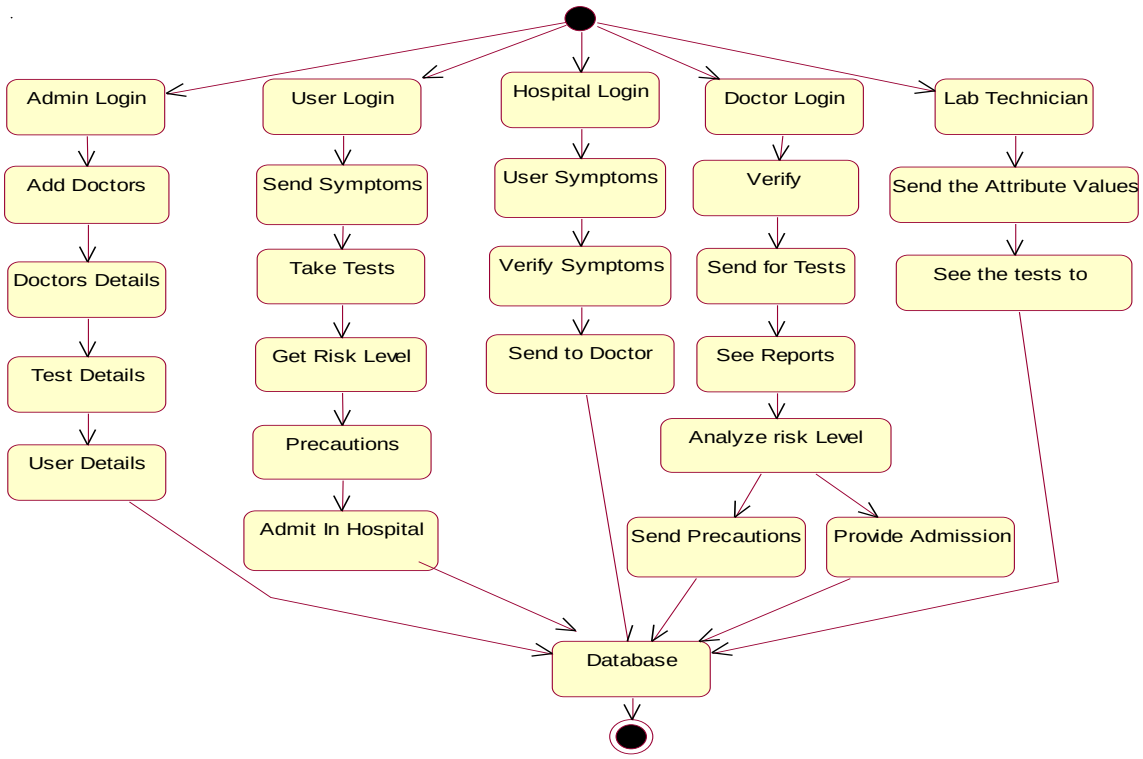
4.3 Object Diagram



EXPLANATION:

In the above digram tells about the flow of objects between the classes. It is a diagram that shows a complete or partial view of the structure of a modeled system. In this object diagram represents how the classes with attributes and methods are linked together to perform the verification with security.

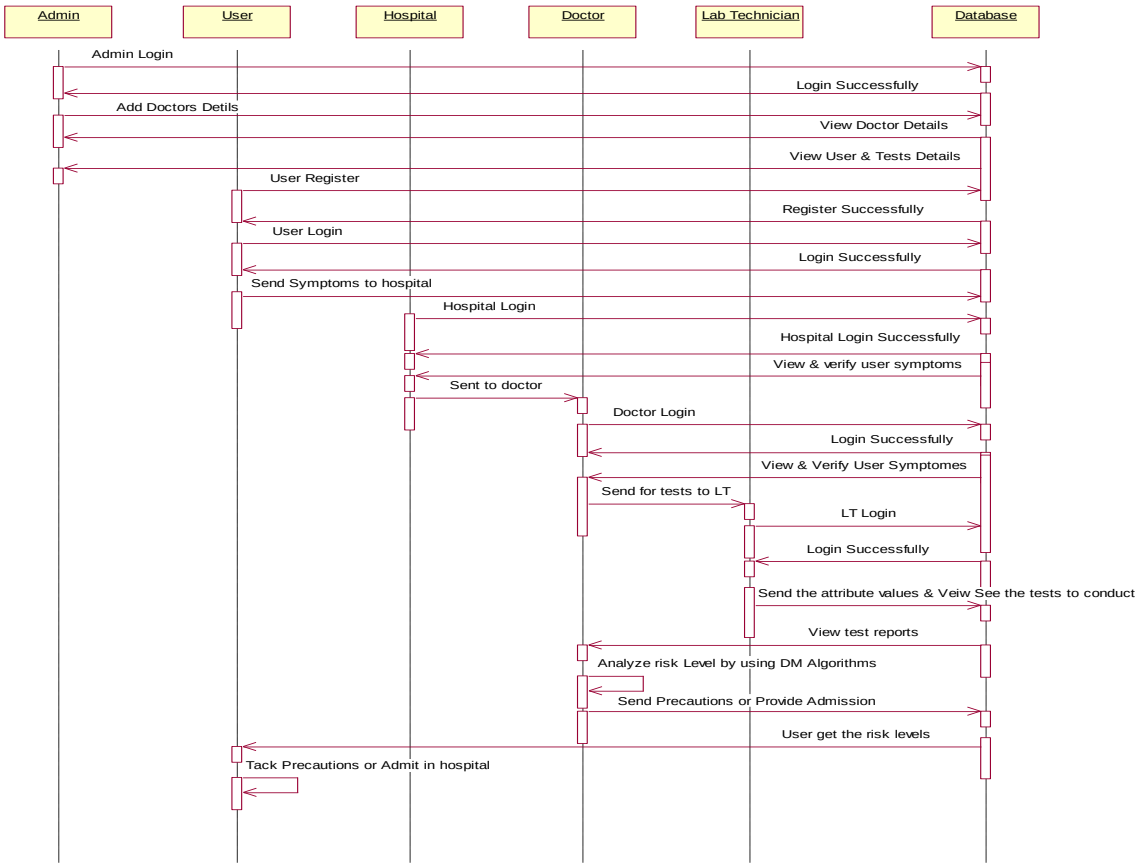
4.4 State Chart Diagram



EXPLANATION:

State diagram are a loosely defined diagram to show workflows of stepwise activities and actions, with support for choice, iteration and concurrency. State diagrams require that the system described is composed of a finite number of states; sometimes, this is indeed the case, while at other times this is a reasonable abstraction. Many forms of state diagrams exist, which differ slightly and have different semantics.

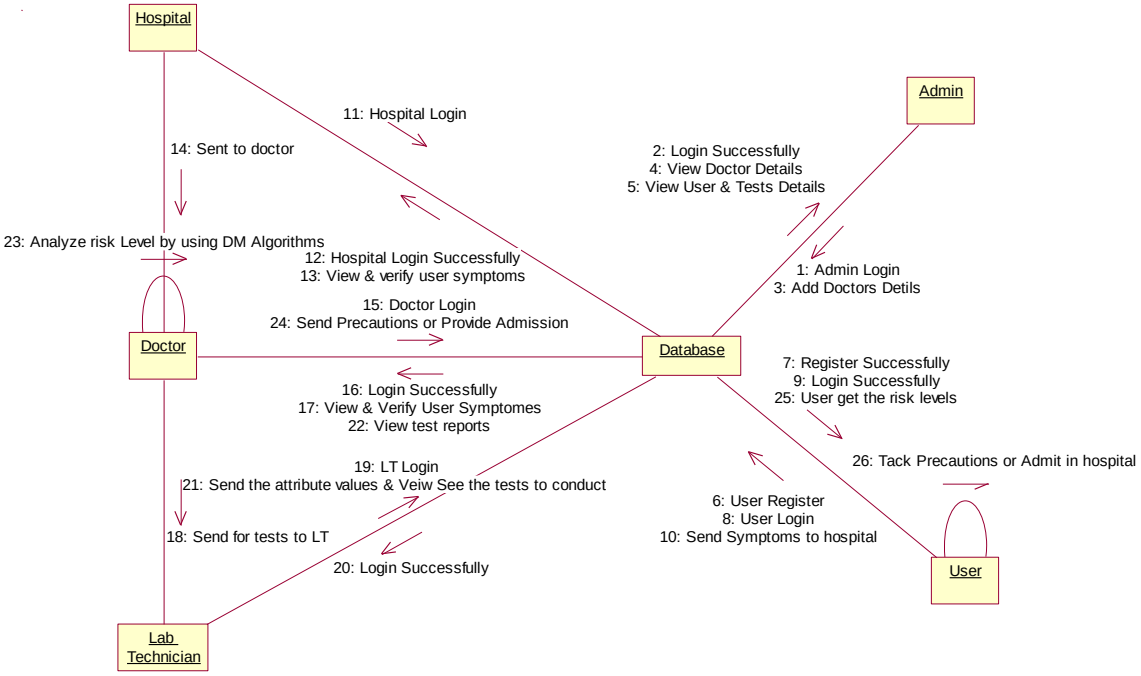
4.5 Sequence Diagram



EXPLANATION:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario.

4.6 Collaboration Diagram



EXPLANATION:

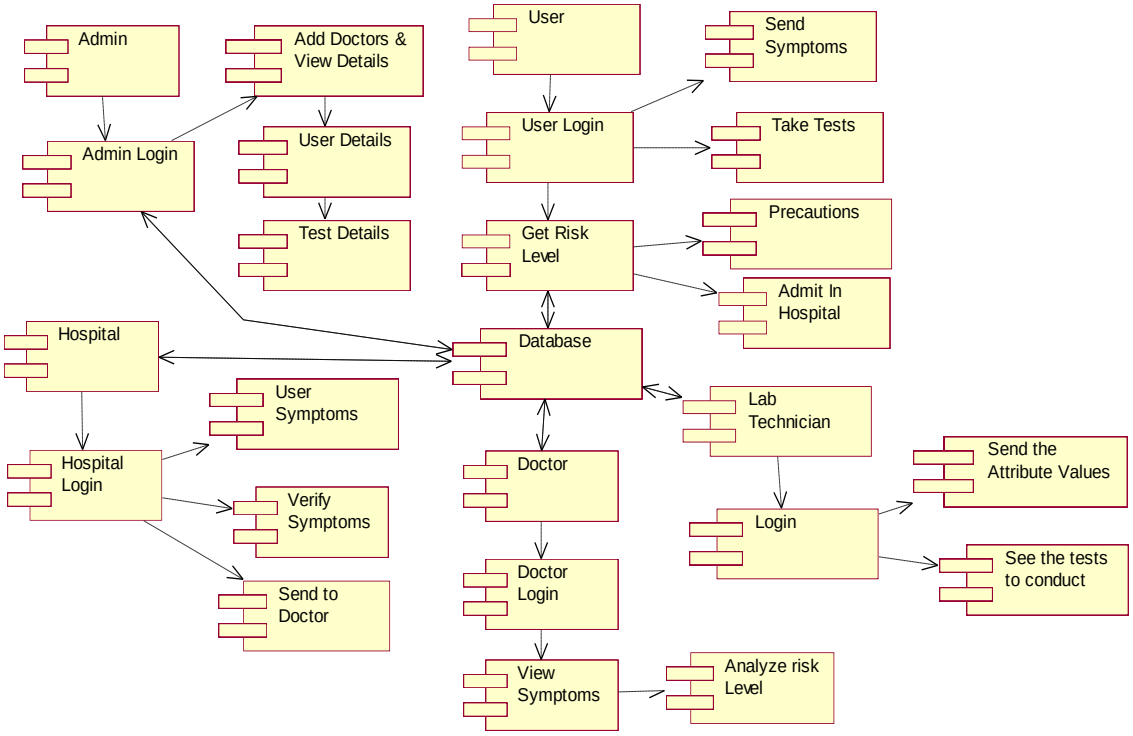
A collaboration diagram, also called a communication diagram or interaction diagram, is an illustration of the relationships and interactions among software objects in the Unified Modeling Language (UML). The concept is more than a decade old although it has been refined as modeling paradigms have evolved.

4.7 Activity Diagram

EXPLANATION:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

4.8 Component Diagram

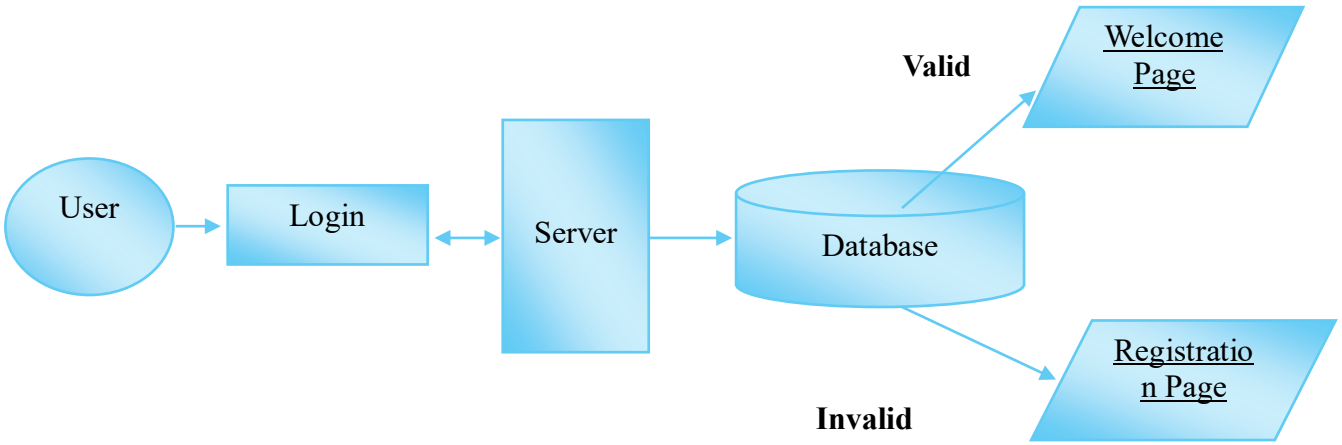


EXPLANATION:

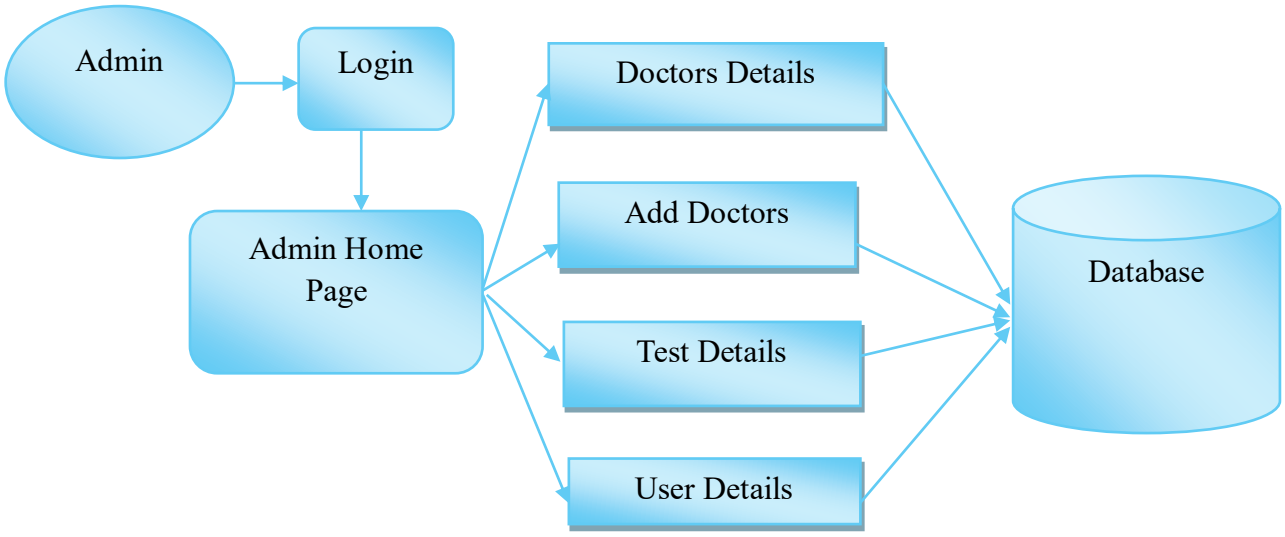
In the Unified Modeling Language, a component diagram depicts how components are wired together to form larger components and or software systems. They are used to illustrate the structure of arbitrarily complex systems. User gives main query and it converted into sub queries and sends through data dissemination to data aggregators. Results are to be showed to user by data aggregators. All boxes are components and arrow indicates dependencies.

4.9 Data Flow Diagram:

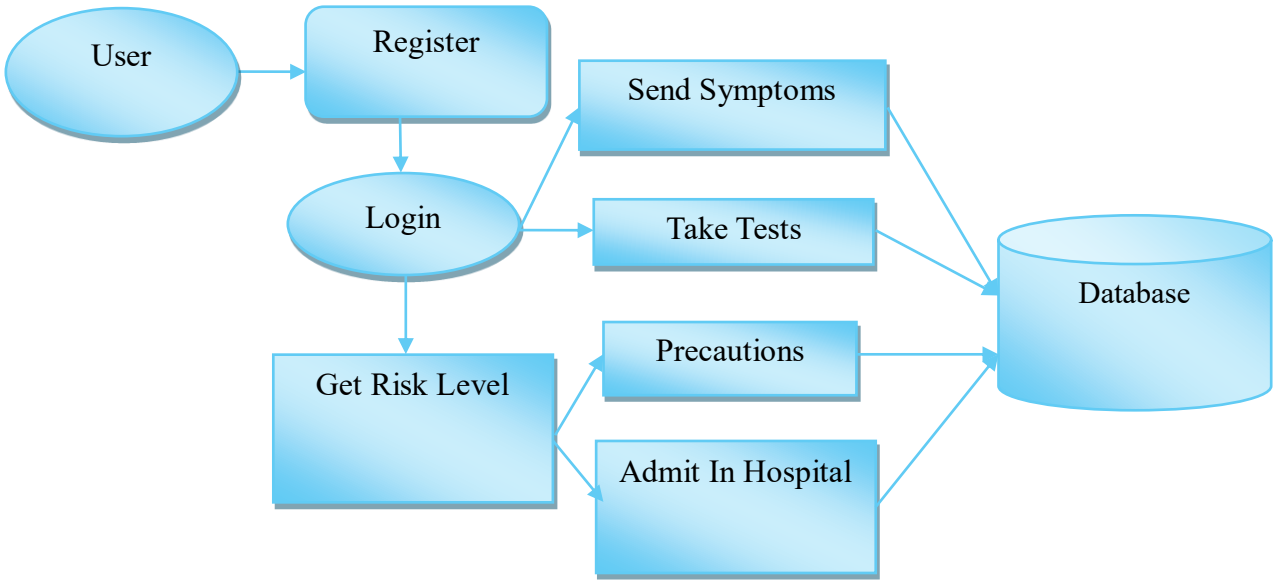
Level 0:



Level 1:



Level 2:

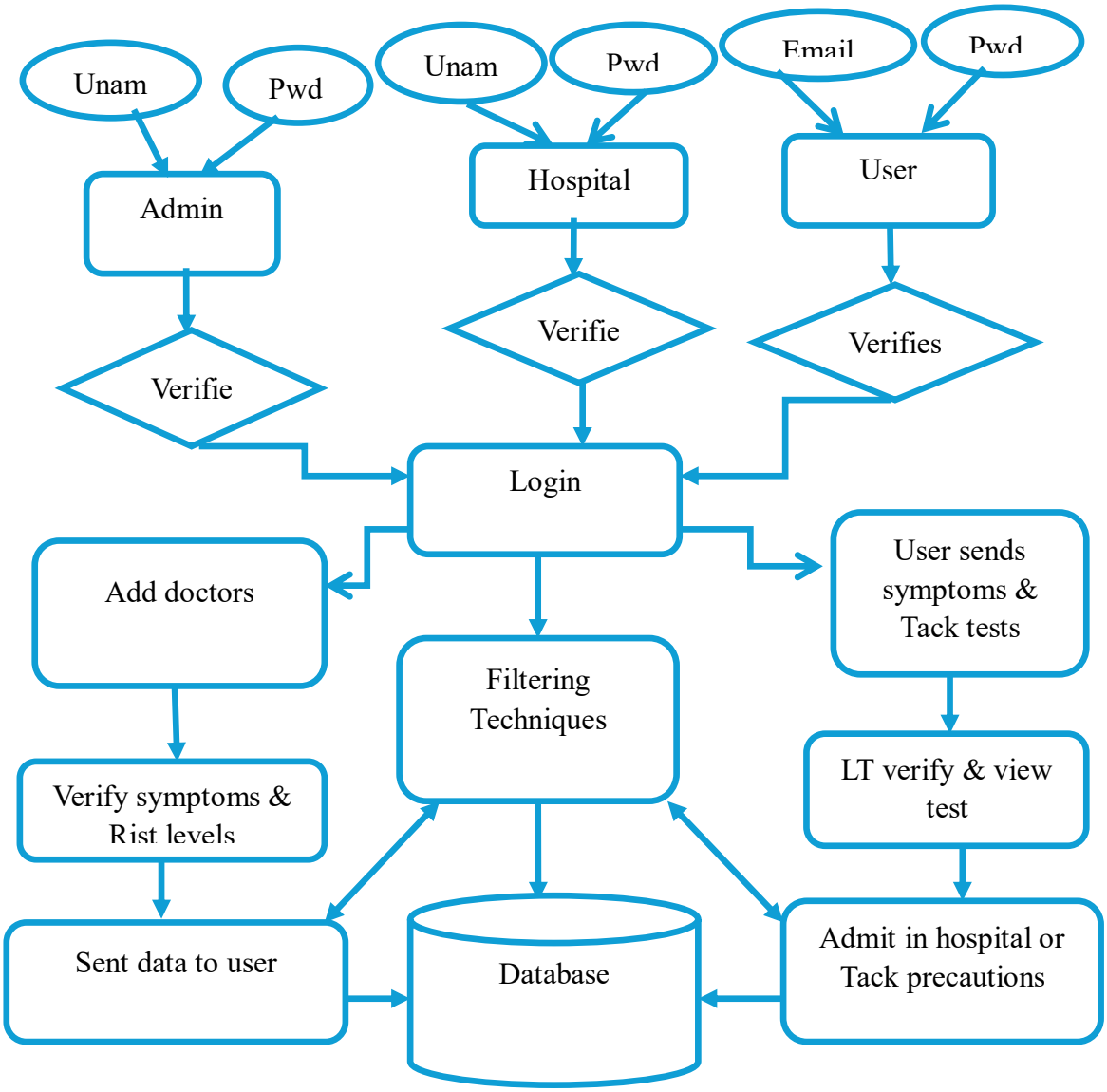


EXPLANATION:

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. Often they are a preliminary step used to create an overview of the system, which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design).

A DFD shows what kinds of data will be input to and output from the system, where the data will come from and go to, and where the data will be stored. It does not show information about the timing of processes, or information about whether processes will operate in sequence or in parallel.

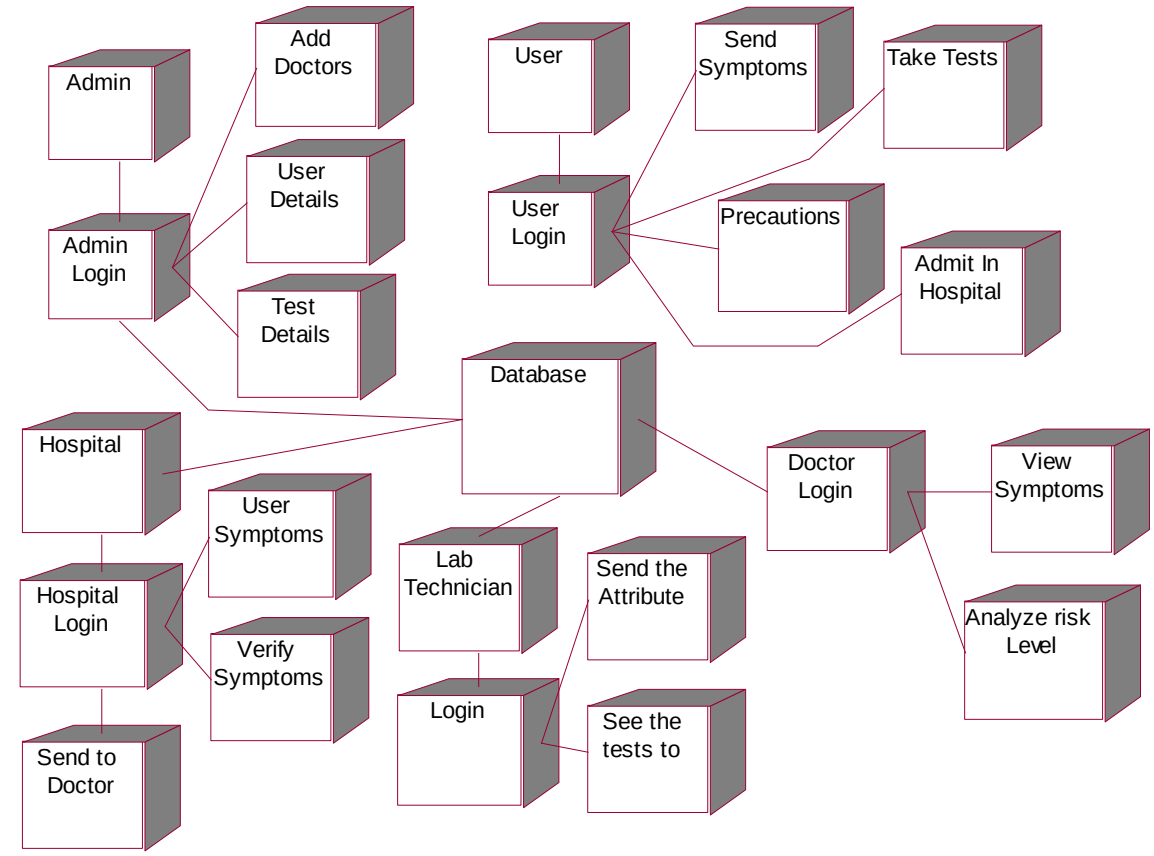
4.10 E-R Diagram:



EXPLANATION:

Entity-Relationship Model (ERM) is an abstract and conceptual representation of data. Entity-relationship modeling is a database modeling method, used to produce a type of conceptual schema or semantic data model of a system, often a relational database.

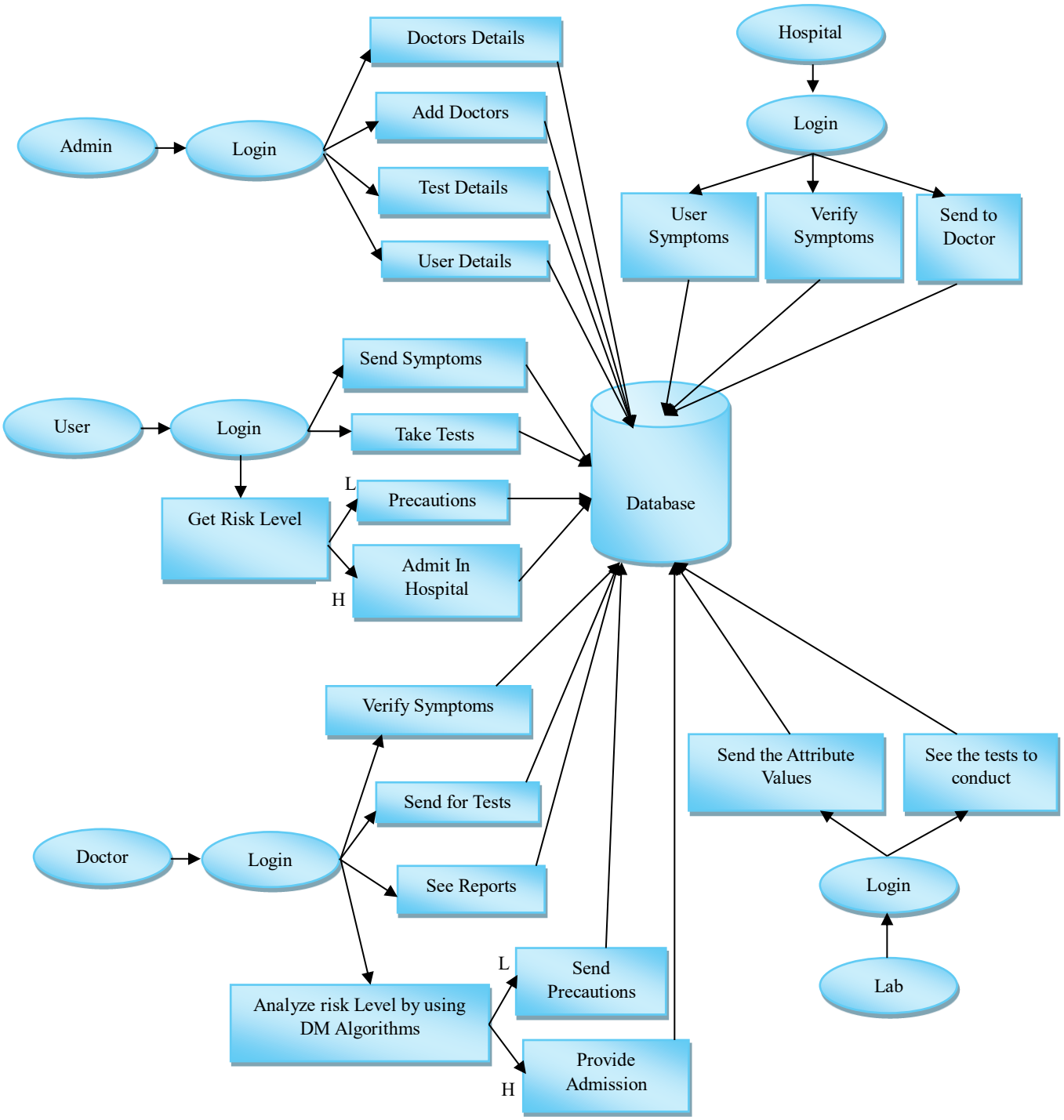
4.11 Deployment Diagram:



EXPLANATION:

In the UML, a component diagram depicts how components are wired together to form larger deployment and or software systems. They are used to illustrate the structure of arbitrarily complex systems. User gives main query and it converted into sub queries and sends through data dissemination. Results are to be showed to user by data aggregators. All boxes are arrow indicates dependencies.

4.12 System Architecture



5. IMPLEMENTATION

5.IMPLEMENTATION

5.1 Administrator Authentication

The Admin.java servlet is responsible for handling the administrator authentication process in the system. It validates the admin's login credentials entered through the web interface. If the credentials match the predefined values, the admin is redirected to the admin home page; otherwise, an error message is displayed. This servlet ensures secure access control, allowing only authorized administrators to manage system functions such as adding doctors, viewing users, and monitoring reports.

Admin.java

```
package com.control;

import java.io.IOException;

import java.io.PrintWriter;

import javax.servlet.ServletException;

import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;

import javax.servlet.http.HttpServletRequest;

import javax.servlet.http.HttpServletResponse;

@WebServlet("/Admin")

public class Admin extends HttpServlet {

private static final long serialVersionUID = 1L;

    public Admin() {`
```

```
    super();

}protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {

                    }    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {

                    PrintWriter o = response.getWriter();

                    String uid = request.getParameter("uid");

                    String pwd = request.getParameter("pwd");

                    if(uid.equals("admin") && pwd.equals("admin")){

                     response.sendRedirect("ahome.jsp");

                    }else{

                     o.println("<script type=\"text/javascript\">");

                     o.println("alert('Please enter valid Details');");

                     o.println("window.location='admin.jsp';</script>");

                    }

}

}
```

5.2 Doctor Registration Functionality

The AddDoc.java servlet manages the **doctor registration functionality** within the healthcare application. It enables the administrator to add new doctor profiles to the system by collecting details such as ID, name, email, password, contact number, and designation. After validating the inputs, it stores the data into the database using the Dao class. Successful entries trigger a confirmation alert, while invalid details prompt an error message. This module simplifies doctor onboarding and centralizes doctor data management.

AddDoc.java

```
package com.control;

import java.io.IOException;

import java.io.PrintWriter;

import javax.servlet.ServletException;

import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;

import javax.servlet.http.HttpServletRequest;

import javax.servlet.http.HttpServletResponse;

import com.bean.DoctorBean;

import com.dao.Dao;

@WebServlet("/AddDoc")

public class AddDoc extends HttpServlet {

private static final long serialVersionUID = 1L;
```

```
    public AddDoc() {

        super();

    }

protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {

                                }

protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {

                                PrintWriter o = response.getWriter();

                                String did = request.getParameter("did");

                                String name = request.getParameter("name");

                                String email = request.getParameter("uid");

                                String pwd = request.getParameter("pwd");

                                String mob = request.getParameter("mob");

                                String des = request.getParameter("des");

                                DoctorBean db = new DoctorBean();

                                db.setDid(did);

                                db.setName(name);

                                db.setEmail(email);

                                db.setPwd(pwd);
```

```
db.setMob(mob);

db.setDes(des);

String sql = "insert into doctors values(?,?,?,?,?,?)";

int i = Dao.addDoctor(sql, db);

if(i > 0){

 o.println("<script type=\"text/javascript\">");

 o.println("alert('Doctor Added Successfully...');");
o.println("window.location='adoctor.jsp';</script>");

}else{   o.println("<script type=\"text/javascript\">");

 o.println("alert('Please enter valid Details');");
o.println("window.location='adddoctor.jsp';</script>");

}

}

}
```

5.3 Doctor Login Authentication

The Doctor.java servlet handles doctor login authentication. It validates the doctor's credentials against records stored in the database and, upon successful verification, establishes a user session and redirects to the doctor's home page. If login fails, an alert message is displayed requesting valid details. This servlet ensures that only registered doctors can access patient information, view medical reports, and provide diagnostic recommendations within the system.

Doctor.java

```
package com.control;

import java.io.IOException;

import java.io.PrintWriter;

import javax.servlet.ServletException;

import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;

import javax.servlet.http.HttpServletRequest;

import javax.servlet.http.HttpServletResponse;

import javax.servlet.http.HttpSession;

import com.dao.Dao;

@WebServlet("/Doctor")

public class Doctor extends HttpServlet {

private static final long serialVersionUID = 1L;
```

```
    public Doctor() {

        super();

    }                                protected void doPost(HttpServletRequest request,
HttpServletResponse response) throws ServletException, IOException {

                                     PrintWriter o = response.getWriter();

                                     String uid = request.getParameter("uid");

                                     String pwd = request.getParameter("pwd");

String sql = "select * from doctors where email='"+uid+"' and password='"+pwd+"'";

                                     HttpSession session = request.getSession();

                                     if(Dao.getLogin(sql) == true){

                                     sql = "select name from doctors where email='"+uid+"'";

                                      String name = Dao.getName(sql);

                                      session.setAttribute("uid", uid);

                                      session.setAttribute("name", name);

                                      response.sendRedirect("dhome.jsp");

                                     }else{

                                      o.println("<script type=\"text/javascript\">");

                                      o.println("alert('Please enter valid Details');");

o.println("window.location='doctors-single.html';</script>");}}}
```

6. TESTING AND DEBUGGING

6.TESTING & DEBUGGING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

6.1 DEVELOPING METHODOLOGIES

The test process is initiated by developing a comprehensive plan to test the general functionality and special features on a variety of platform combinations. Strict quality control procedures are used. The process verifies that the application meets the requirements specified in the system requirements document and is bug free. The following are the considerations used to develop the framework from developing the testing methodologies.

6.2 Unit Testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program input produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

6.3 Functional Test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals. Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

6.4 System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

6.5 Performance Test

The Performance test ensures that the output be produced within the time limits, and the time taken by the system for compiling, giving response to the users and request being send to the system for to retrieve the results.

7. RESULT & DISCUSSION

7.RESULT & DISCUSSION

This project is implements like web application using COREJAVA and the Server process is maintained Cascading Style Sheet plays using the SOCKET & SERVERSOCKET and the Design part.

7.1 VARIOUS SNAPSHOTS

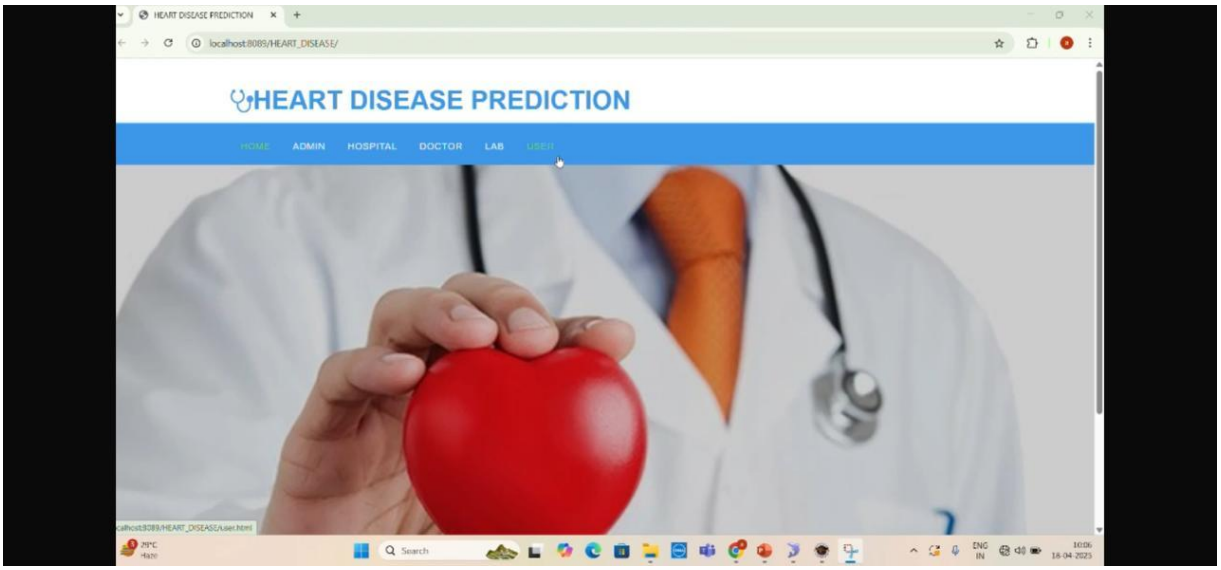


FIGURE 8: HOME PAGE

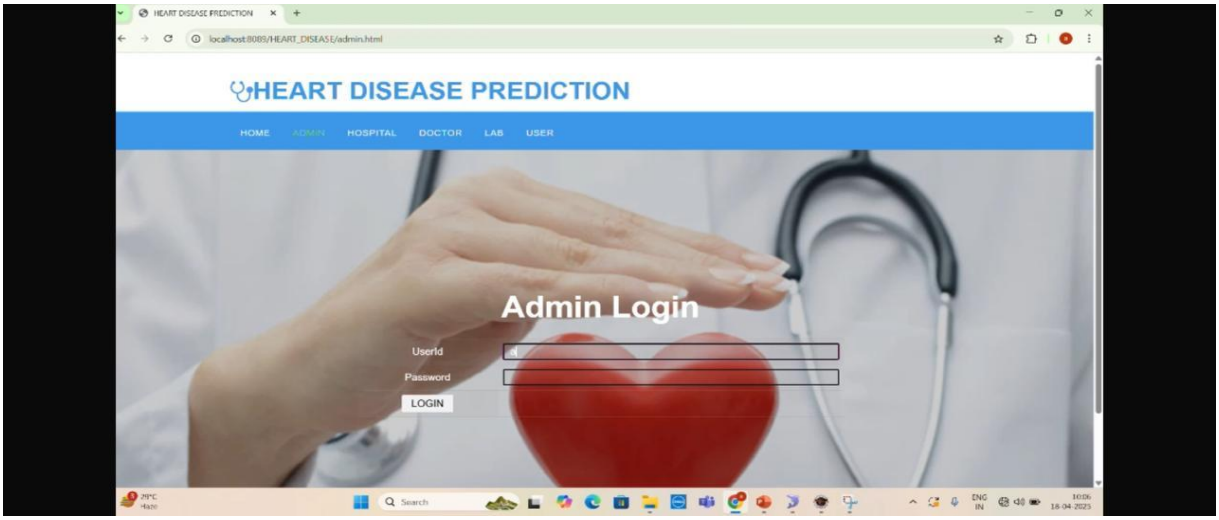


FIGURE 9: ADMIN LOGIN

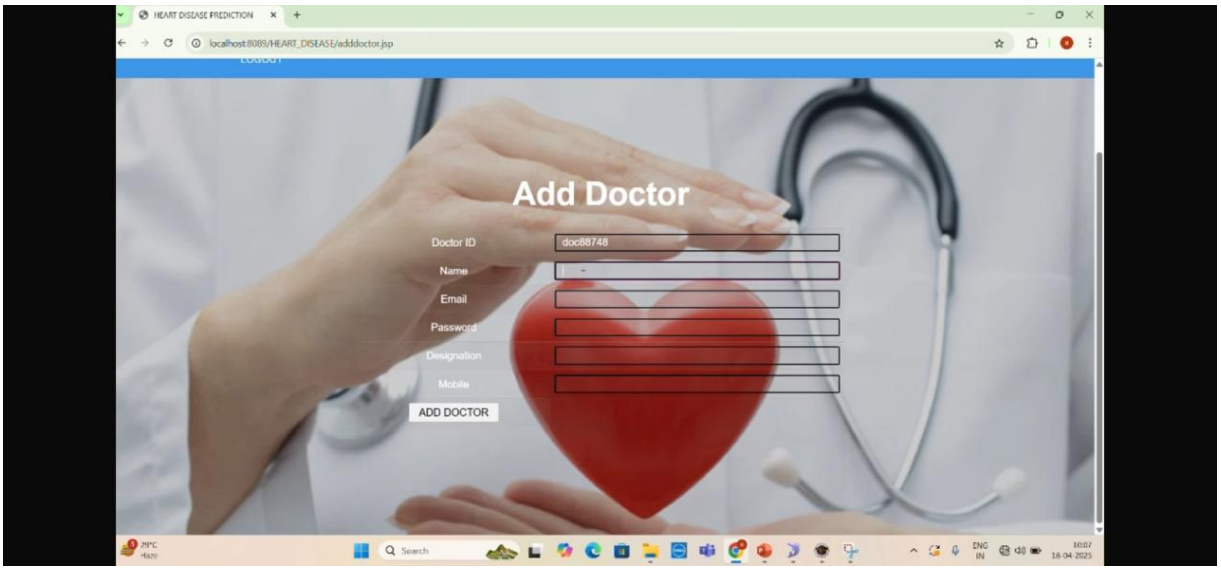


FIGURE 10: ADD DOCTOR

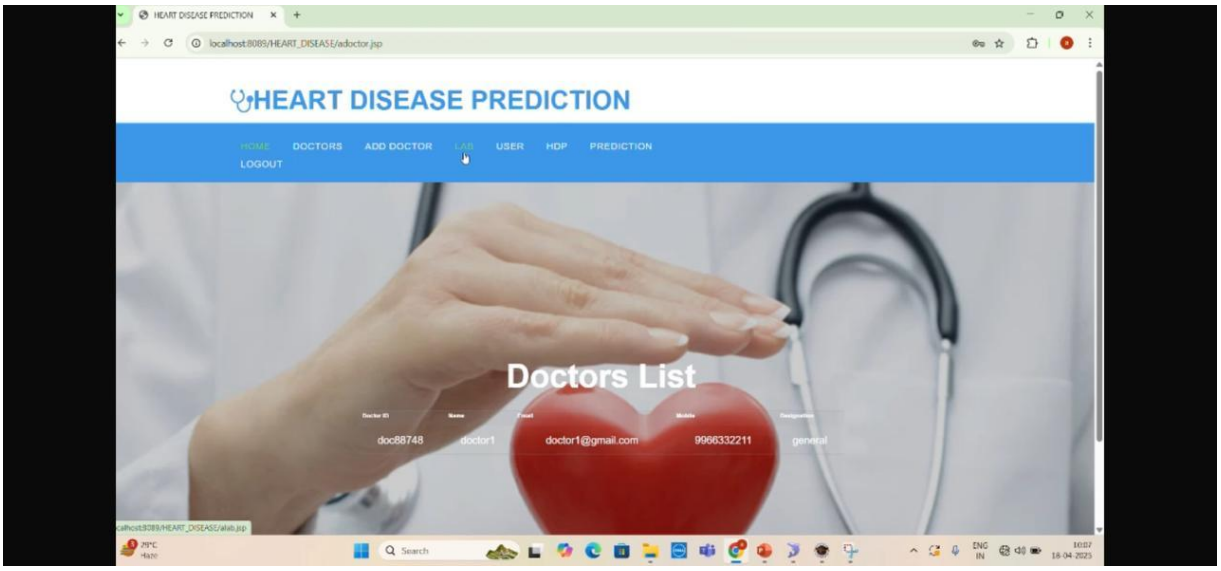


FIGURE 11: DOCTOR LIST

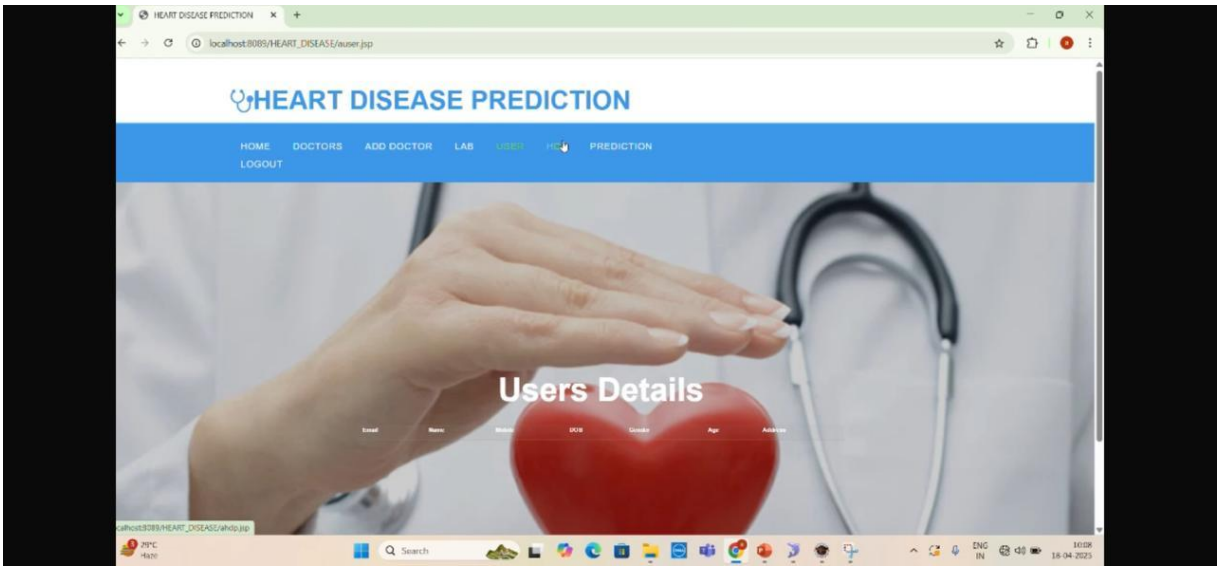


FIGURE 12: USER DETAILS

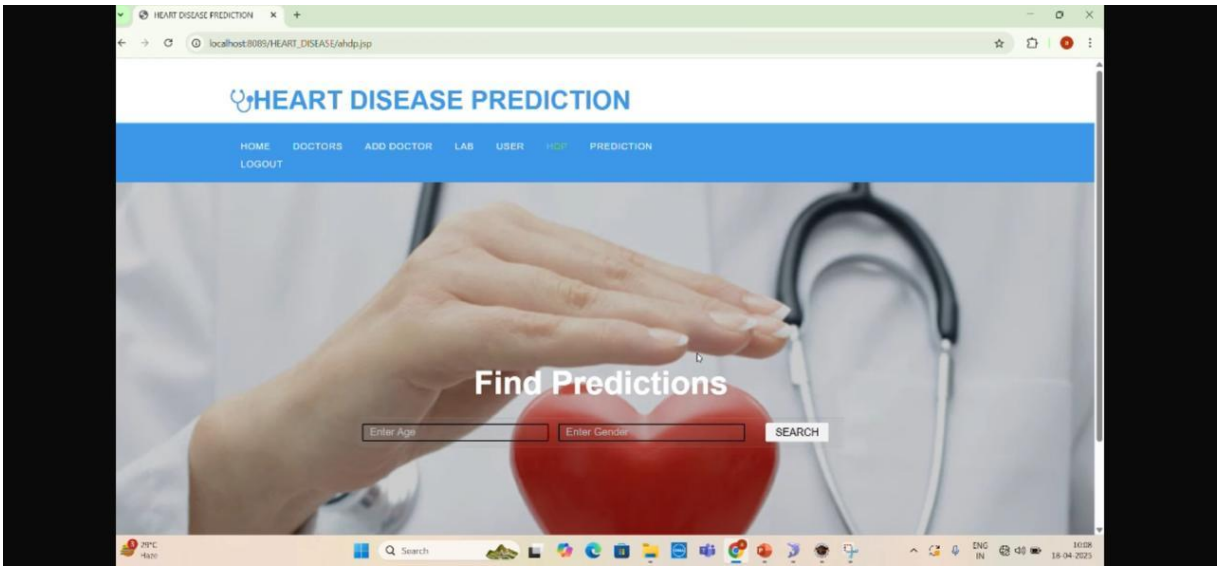


FIGURE 13: PREDICTION OF DISEASE

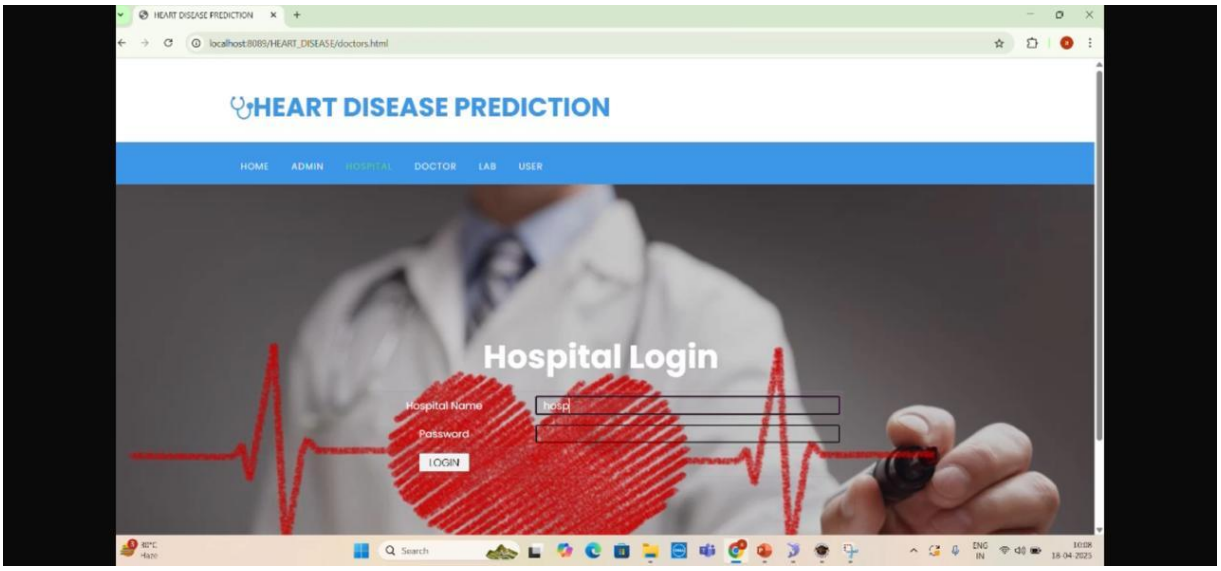


FIGURE 14: HOSPITAL LOGIN

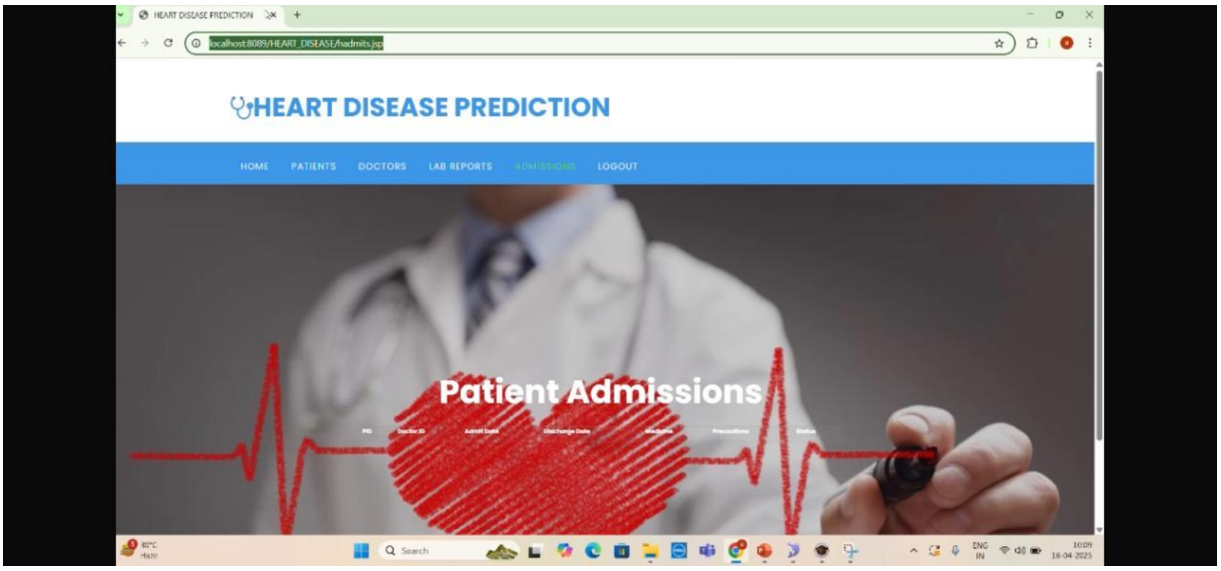


FIGURE 15: PATIENT ADMISSION

FIGURE 16: DOCTORS LOGIN

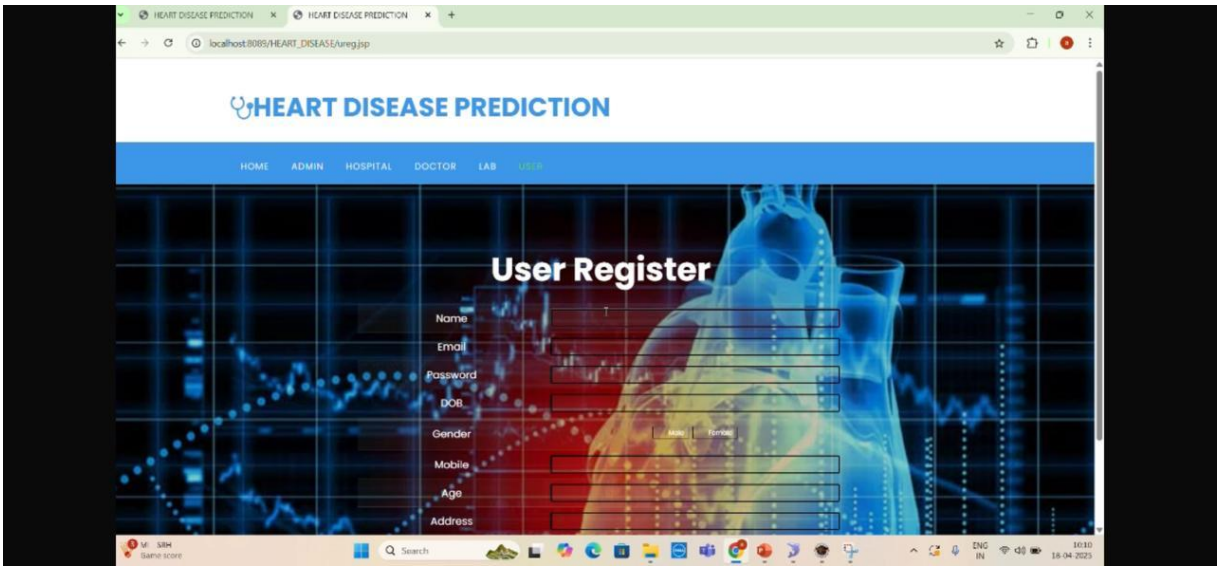


FIGURE 17: USER REGISTRATION

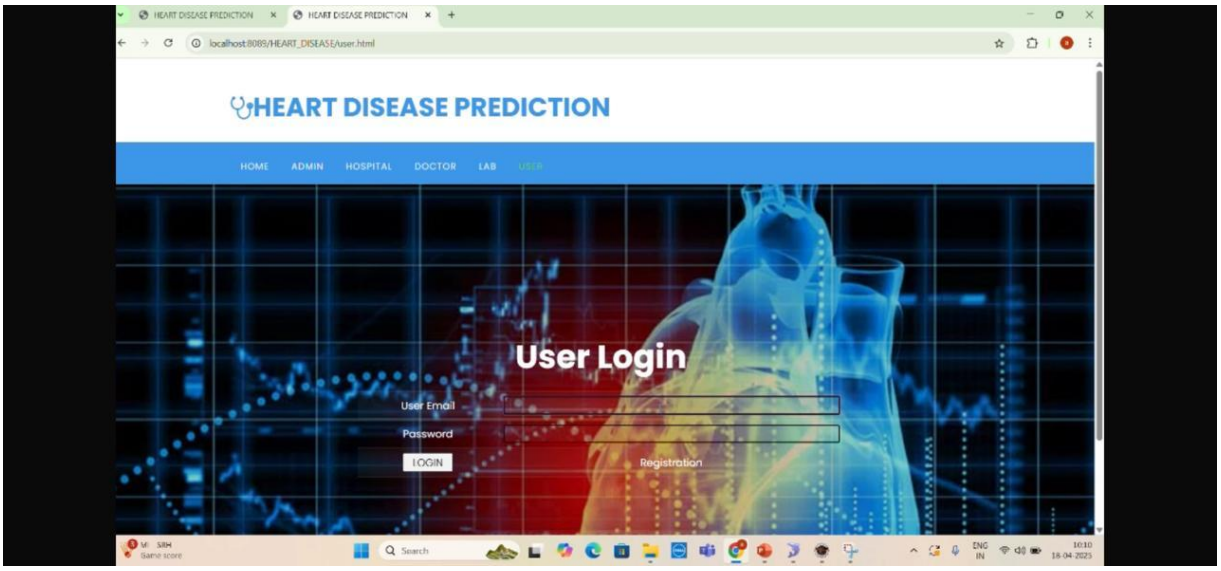


FIGURE 18: USER LOGIN

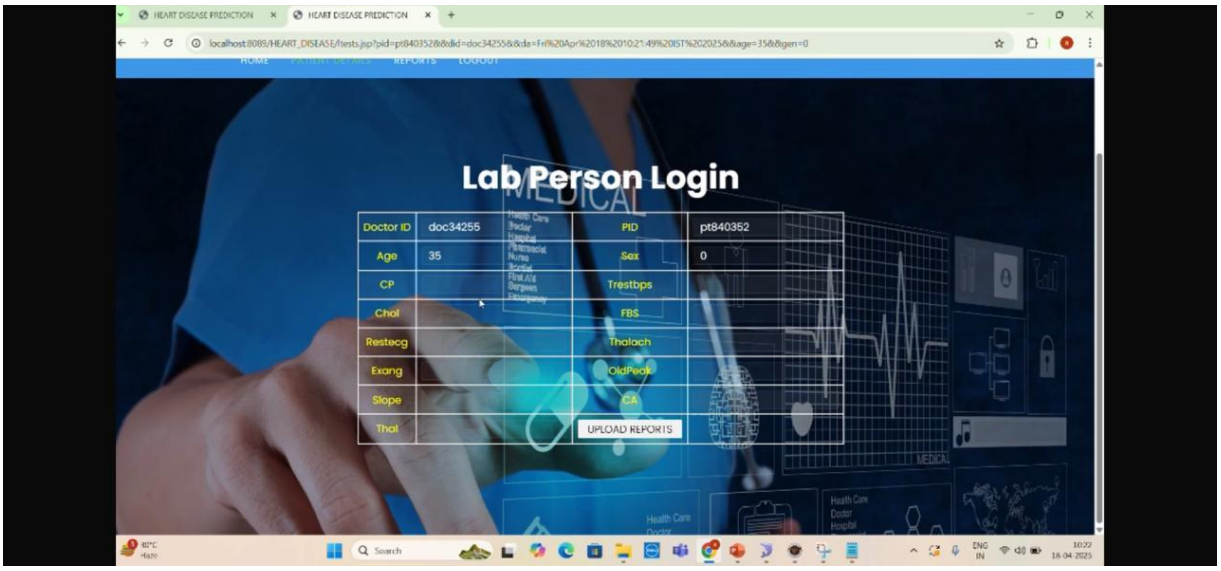


FIGURE 19: LAB PERSON LOGIN FOR PREDICTION DETAILS

8. CONCLUSION

8.CONCLUSION

In this study, we provide a novel neural network method to estimate the status of CHD. By applying a scrutiny mechanism with EOA to obtain an adaptive weight for the hybrid method, the model extracts unique and common embedding from topology, node attributes, and their collaborations with our experimental testing on the dataset. The O-SBGC LSTM acquires the most essential information from the node and improves model accuracy by a large margin. The experiment also proves that the proposed method exhibits better results in various performance metrics than several existing baselines. To improve the prediction performance, the excellent prediction ability will optimize its application in the diagnosis and treatment of postoperative recurrence while simplifying the diagnosis process with an accuracy of 98%. The proposed work also manages to prevent CHD by providing suggestions according to the CHD disease level. Fuzzy inference systems take different clinical data and generate decisions for doctors for effective treatment, so that the risk to the patient can be avoided.

9. BIBLIOGRAPHY

9. BIBLIOGRAPHY

1. P. Saeedi, I. Petersohn, P. Salpea, B. Malanda, S. Karuranga, N. Unwin, et al., "Global and regional diabetes prevalence estimates for 2019 and projections for 2030 and 2045: Results from the international diabetes federation diabetes atlas 9th edition", Diabetes Res. Clin. Pract., vol. 157, Nov. 2019.

2. M. Sachdev and A. Misra, "Heterogeneity of dietary practices in India: Current status and implications for the prevention and control of type 2 diabetes", Eur. J. Clin. Nutrition, vol. 77, no. 2, pp. 145-155, 2023.

3. Ministry of Health and Family Welfare, New Delhi, India, 2016.

4. B. J. Gersh, "Incidence of cardiovascular risk factors in an Indian urban cohort: Results from the New Delhi Birth Cohort", Yearbook Cardiol., vol. 2012, pp. 249-252, Jan. 2012.

5. F. B. Hu, "Globalization of diabetes: The role of diet lifestyle and genes", Diabetes care, vol. 34, no. 6, pp. 1249-1257, 2011.

6. K. Dhana, J. Nano, S. Ligthart, A. Peeters, A. Hofman, W. Nusselder, et al., "Obesity and life expectancy with and without diabetes in adults aged 55 years and older in The Netherlands: A prospective cohort study", PLOS Med., vol. 13, no. 7, Jul. 2016.

7. H. Y. Yong, Z. M. Shariff, B. N. M. Yusof, Z. Rejali, Y. Y. S. Tee, J. Bindels, et al., "Independent and combined effects of age body mass index and gestational weight gain on the risk of gestational diabetes mellitus", Sci. Rep., vol. 10, no. 1, pp. 8486, May 2020.

8. V. Mohan, G. Radhika, P. Vijayalakshmi and V. Sudha, "Can the diabetes/cardiovascular disease epidemic in India be explained at least in part by excess refined grain (rice) intake?", Indian J. Med. Res., vol. 131, no. 3, pp. 369-372, 2010.

9. L. A. Zilliox, K. Chadrasekaran, J. Y. Kwan and J. W. Russell, "Diabetes and cognitive impairment", Current Diabetes Rep., vol. 16, no. 2, pp. 1-11, 2016.

10. A. H. Gonsalves, F. Thabtah, R. M. A. Mohammad and G. Singh, "Prediction of coronary heart disease using machine learning: An experimental analysis", Proc. 3rd Int. Conf. Deep Learn. Technol., pp. 51-56, Jul. 2019.

11. T. Nazir, A. Irtaza, A. Javed, H. Malik, D. Hussain and R. A. Naqvi, "Retinal image analysis for diabetes-based eye disease detection using deep learning", Appl. Sci., vol. 10, no. 18, pp. 6185, Sep. 2020.

12. J. Nahar, T. Imam, K. S. Tickle and Y.-P.-P. Chen, "Association rule mining to detect factors which contribute to heart disease in males and females", Expert Syst. Appl., vol. 40, no. 4, pp. 1086-1093, Mar. 2013.